

MINISTRY OF EDUCATION & TRAINING

STATE BANK OF VIETNAM

**HO CHI MINH UNIVERSITY OF BANKING**



**BACHELOR THESIS**

**THE FACTORS INFLUENCING THE FINANCIAL  
SUSTAINABILITY OF VIETNAMESE COMMERCIAL BANKS:  
A QUANTILE REGRESSION ANALYSIS**

MAJOR: FINANCE AND BANKING

CODE: 7 34 02 01

**LAM THUY KIEU**

**HO CHI MINH CITY, 2025**

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**HO CHI MINH CITY, 2025**

## ABSTRACT

This study investigates the factors influencing the financial sustainability of Vietnamese commercial banks from 2010 to 2023. By employing the System Generalized Method of Moments (SGMM) and quantile regression, the research provides an in-depth analysis of the effects of both bank-specific characteristics and macroeconomic variables on financial sustainability. The empirical findings indicate that return on assets (ROA), operating income (lnINC), loan-to-deposit ratio (LTD), board independence (BI), and economic growth (GDP) have a positive impact on financial sustainability. Conversely, bank size (lnSIZE), non-performing loans (NPL), and inflation (INF) negatively affect sustainability. Meanwhile, liquidity (LIQ) and capital adequacy ratio (CAR) do not show statistically significant effects. These findings highlight the importance of efficient risk management, governance structures, and macroeconomic stability in ensuring the long-term financial sustainability of banks. The study contributes to the existing literature by offering empirical evidence on the financial sustainability of banks in an emerging market context. It also provides practical implications for policymakers and banking executives in developing strategies to enhance financial sustainability and resilience in the Vietnamese banking sector.

**Keywords:** *Commercial banks, financial sustainability, SGMM, quantile regression.*

## TÓM TẮT

Nghiên cứu này điều tra các yếu tố ảnh hưởng đến sự bền vững tài chính của các ngân hàng thương mại Việt Nam trong giai đoạn 2010–2023. Bằng cách áp dụng phương pháp SGMM và hồi quy phân vị, nghiên cứu cung cấp phân tích chuyên sâu về tác động của cả các đặc điểm riêng của ngân hàng và các biến vĩ mô đối với sự bền vững tài chính. Dữ liệu được thu thập từ 27 Ngân hàng TMCP Việt Nam với 378 quan sát trong 14 năm

Kết quả thực nghiệm cho thấy tỷ suất lợi nhuận trên tài sản (ROA), thu nhập từ hoạt động (lnINC), tỷ lệ cho vay trên tiền gửi (LTD), tính độc lập của hội đồng quản trị (BI) và tăng trưởng kinh tế (GDP) có tác động tích cực đến sự bền vững tài chính. Ngược lại, quy mô ngân hàng (lnSIZE), tỷ lệ nợ xấu (NPL) và lạm phát (INF) có ảnh hưởng tiêu cực đến sự bền vững. Trong khi đó, tính thanh khoản (LIQ) và tỷ lệ an toàn vốn (CAR) không cho thấy ảnh hưởng có ý nghĩa thống kê.

Những phát hiện này nhấn mạnh tầm quan trọng của quản lý rủi ro hiệu quả, cơ cấu quản trị và ổn định kinh tế vĩ mô trong việc đảm bảo sự bền vững tài chính dài hạn của các ngân hàng. Nghiên cứu này đóng góp vào tài liệu hiện có bằng cách cung cấp bằng chứng thực nghiệm về sự bền vững tài chính của các ngân hàng trong bối cảnh thị trường mới nổi. Đồng thời, nghiên cứu cũng đưa ra các hàm ý thực tiễn cho các nhà hoạch định chính sách và lãnh đạo ngân hàng trong việc xây dựng chiến lược nhằm nâng cao sự bền vững và khả năng chống chịu của hệ thống ngân hàng Việt Nam.

**Từ khóa:** Ngân hàng thương mại, bền vững tài chính, SGMM, hồi quy phân vị.

## AUTHOR'S DECLARATION

I honestly declare that the thesis titled *"The factors influencing the financial sustainability Of Vietnamese Commercial Banks: A quantile regression analysis"* is the result of my accumulated knowledge and application of learned concepts in conducting this research, carried out with the integrity.

This thesis is researched by myself and it represents my dedication and passion throughout my academic journey at the university. The research results are truthful, and there is no previously published content or work conducted by others that has been included in my thesis, except for properly cited references within the thesis.

Ho Chi Minh city, 2025

**Author**

A handwritten signature in black ink, appearing to read 'LAM THUY KIEU', with a long horizontal stroke extending to the right.

LAM THUY KIEU

## ACKNOWLEDGMENT

The completion of this graduation thesis marks a significant milestone in my academic journey. I would like to express my deepest gratitude to the Banking University of Ho Chi Minh City for providing me with the invaluable opportunity to apply theoretical knowledge to practical research.

In particular, I extend my sincerest and most profound appreciation to Dr. Nguyễn Minh Nhật, my thesis advisor, who has always been patient, supportive, and dedicated throughout the research process. His close guidance and invaluable feedback have helped me overcome challenges and successfully complete this study. I am fully aware that, given my limited experience and knowledge, this thesis is unlikely to be free from shortcomings. Therefore, I sincerely welcome any constructive feedback from the esteemed faculty of the Banking University of Ho Chi Minh City, as well as further guidance from Dr. Nguyễn Minh Nhật, to refine and improve my research in the future.

Finally, I would like to extend my best wishes to the faculty members and staff of the Banking University of Ho Chi Minh City for continued good health and success in their noble mission of education. To Dr. Nguyễn Minh Nhật, I wish you health, happiness, and continued achievements in your research and teaching career.

I sincerely thank you!

Ho Chi Minh city, 2025

**Author**



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## LIST OF ABBREVIATIONS

| ABBREVIATIONS | MEANING                                                                                                 |
|---------------|---------------------------------------------------------------------------------------------------------|
| BI            | Board Independence Ratio                                                                                |
| BOD           | Board Of Directors                                                                                      |
| CAMELS        | Capital Adequacy, Asset Quality, Management Quality, Earnings, Liquidity and Sensitivity to Market Risk |
| CAR           | Capital Adequacy Ratio                                                                                  |
| CSR           | Corporate Social Responsibility                                                                         |
| ESG           | Environmental, Social, and Governance                                                                   |
| FEM           | Fixed Effects Model                                                                                     |
| FGLS          | Feasible Generalized Least Squares                                                                      |
| FSR           | Financial sustainability of Commercial Banks                                                            |
| GDP           | Economic Growth                                                                                         |
| INF           | Inflation rate                                                                                          |
| LIQ           | Liquidity Ratio                                                                                         |
| lnINC         | Operating Income                                                                                        |
| lnSIZE        | Bank size                                                                                               |
| LTD           | Loan to Deposit Ratio                                                                                   |
| MFIs          | Micro - Finance institutions                                                                            |

|      |                                  |
|------|----------------------------------|
| NPL  | Non performing Loan              |
| OLS  | Ordinary Least Squares           |
| OLS  | Ordinary Least Squares           |
| REM  | Random Effects Model             |
| ROA  | Return on assets                 |
| SGMM | System General Method of Moments |

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## **CHAPTER 1: INTRODUCTION**

### **1.1. RATIONALE OF THE RESEARCH**

The appearance of financial crises over the past decades has increased concerns regarding the financial sustainability of banks. After the 2008 global economic crisis, banks faced a lot of challenges, including non-performing loans, declining profitability, financial losses, restructuring, and other difficulties. The international connection of banks through stock markets and lending activities has caused a chain of consequences. Consequently, the Vietnamese economy and banking system were not immune to these adverse impacts.

Then, the outbreak of the COVID-19 pandemic and systemic risks have manifested in several aspects, such as disruptions in direct banking operations, excessive credit growth, adjustments in interest rate exemption and reduction policies, debt restructuring while maintaining debt classification, and severe liquidity crises. Recently, geopolitical conflicts, particularly the Russia-Ukraine war and other regional disputes, have posed threats to Vietnam's banking system such as increasing financial and credit risks, as well as exposing banks to heightened foreign exchange risks due to fluctuations in global financial markets. During this period, it is highly important for banks to enhance financial supervision, adapt to volatile investment environments, comprehensively manage risks, and implement sound financial strategies to ensure sustainability in the current uncertain business climate.

Vietnamese commercial banks play a pivotal role in the financial sector and the broader economy. Banking activities have gradually expanded and diversified, with financial stability and sustainability in banking operations being crucial for the long-term growth and resilience of Vietnam's financial system and economy. The financial sustainability of individual commercial banks forms the foundation for the stability of the entire financial system. The case of Sai Gon Joint Stock Commercial Bank (SCB) in late 2022, despite being mitigated by timely intervention from the State Bank of Vietnam (SBV), had notable repercussions on the banking sector and

the economy. This incident underscored the necessity of strengthening risk management, regulatory oversight, and governance mechanisms to maintain the stability and sustainable development of the banking system, both at the individual bank level and across the entire Vietnamese banking sector.

There are several studies about financial sustainability in banking globally. Some research suggests that financial sustainability is influenced by income diversification from interest and non-interest income sources, as evidenced in studies by Köhler (2015) and Mergaerts (2016). However, Köhler (2015) also highlighted that certain banks face greater risks when diversifying their income streams. Other studies have focused on factors such as bank size, efficiency, and leverage, which can impact financial sustainability, as discussed by Albaity et al. (2019).

In Vietnam, existing researches primarily concentrated on bank performance, such as the studies by Nguyen & Pham (2020), or the financial stability of Vietnamese joint-stock commercial banks, as explored by Le Ngoc Quynh Anh (2020) and Nguyen Duy Quoc (2020). A general limitation of these studies is their emphasis on short-term operational efficiency rather than a comprehensive assessment of the factors affecting financial sustainability. Overall, domestic research has predominantly focused on evaluating the performance of commercial banks, with limited studies addressing the long-term sustainability of the banking sector.

Based on acquired knowledge and extensive research, this study seeks to provide additional empirical evidence on the factors influencing the financial sustainability of Vietnam's commercial banking system. The research topic "The factors influencing the financial sustainability of Vietnamese Commercial Banks: A quantile regression analysis" has been chosen to analyze the determinants of financial sustainability, identify risk factors, and propose measures to enhance banks' resilience, thereby contributing to financial system stability and preventing financial crises. The study's findings offer empirical insights for policymakers in formulating and refining banking supervision and regulatory policies. Regulatory authorities can

utilize these results to adjust policies appropriately, aiming for the sustainable development of the banking system.

## **1.2. RESEARCH OBJECTIVES**

### ***1.2.1. General objectives***

The general objective of this study is to accurately assess and analyze the impact direction of each factor influencing the financial sustainability of Vietnamese commercial banks. Based on these findings, the study proposes recommendations and solutions to help commercial banks develop appropriate strategies, enhance sustainability, and adopt flexible measures to respond to economic fluctuations.

### ***1.2.2. Specific objectives***

Identify the factors influencing the financial sustainability of Vietnamese commercial banks.

Assess the impact level of each factor on the financial sustainability of Vietnamese commercial banks using the quantile regression method.

Provide policy implications and recommendations to enhance the financial sustainability of Vietnamese commercial banks.

## **1.3. RESEARCH QUESTIONS**

Question 1: What factors influence the financial sustainability of Vietnamese commercial banks?

Question 2: To what extent does each factor impact financial sustainability across different quantiles?

Question 3: What policy recommendations and strategic measures can be implemented to improve the financial sustainability of Vietnamese commercial banks?

## **1.4. RESEARCH SUBJECT AND SCOPE**

### ***1.4.1. Research subjects***

The research subject of this study is the factors influencing the financial sustainability of Vietnamese commercial banks.

### ***1.4.2. Research scope***

#### **Spatial scope**

The study analyzes the impact of various factors on the financial sustainability of Vietnamese commercial banks. The research sample consists of financial data and business performance reports from 27 Vietnamese commercial banks, excluding those lacking sufficient data or no longer in operation.

The banks included in the study are: An Binh Commercial Joint Stock; Asia Commercial Joint Stock; Vietnam Bank for Agriculture and Rural Development; Bac A Commercial Joint Stock Bank; Bank for Investment and Development of Vietnam; BV Commercial Joint Stock Bank; Vietnam Joint Stock Commercial Bank for Industry and Trade; Vietnam Export-Import Commercial Joint Stock Bank; Ho Chi Minh City Development Joint Stock Commercial; Kien Long Commercial Joint Stock Bank; Fortune Vietnam Joint Stock Commercial Bank; Petrolimex Group Commercial Joint Stock Bank; Military Commercial Joint Stock Bank; Vietnam Maritime Commercial Joint Stock Bank; Nam A Commercial Joint Stock; National Citizen Commercial Joint Stock Bank; Orient Commercial Joint Stock Bank; Saigon Commercial Joint Stock Bank; Saigon-Hanoi Commercial Joint Stock Bank; Southeast Asia Commercial Joint Stock Bank; Saigon Thuong Tin Commercial Joint Stock Bank; Vietnam Technological and Commercial Joint Stock Bank; Tien Phong Commercial Joint Stock Bank; Vietnam Asia Commercial Joint Stock Bank; Vietnam Prosperity Joint Stock Commercial Bank; Joint Stock Commercial Bank for Foreign Trade of Vietnam; Vietnam International Commercial Joint Stock Bank

#### **Time scope**

The research covers the period from 2010 to 2023, during which Vietnam's banking system transitioned from the post-2008 global financial crisis recovery phase to a stable growth trajectory. This period provides a strong foundation for analyzing factors affecting bank performance. Additionally, from 2010 onward, the financial and operational reports of commercial banks have been fully disclosed with transparency, ensuring data accuracy and reliability for the study.

## **1.5. RESEARCH METHOD**

To estimate the model and evaluate the factors affecting the financial sustainability of Vietnamese commercial banks, this study employs three methods for assessment.

The first stage of this study involves data collection. To ensure the comprehensiveness and reliability of the dataset, the author employed secondary data collection methods by gathering micro and macro data from the financial statements of 27 Vietnamese commercial banks. These financial statements were publicly available on official websites during the period from 2010 to 2023. Additionally, supplementary macroeconomic data were extracted from the World Bank database.

Subsequently, a qualitative research approach was employed to conduct a systematic review, synthesis, and analysis of prior studies, both domestic and international, concerning the factors influencing capital structure in commercial banks. Then, the author was able to construct an appropriate research model that serves as the foundation for further empirical analysis.

Following the model development, the study utilized a quantitative research approach, employing STATA 17 software to conduct preliminary descriptive statistics. This included summarizing key characteristics of the dataset, such as mean, maximum, minimum, and standard deviation values of both dependent and independent variables. The study then proceeded with regression analysis, applying conventional econometric models. In cases where endogeneity issues were detected, the System Generalized Method of Moments (SGMM) estimator, as proposed by Blundell and Bond (1998), was utilized to address potential biases and ensure robust estimation results.

Finally, author performed quantile regression with the `qreg` command to replace OLS regression to assess the impact of factors on each quantile instead of the mean of the variable or when the data distribution is skewed.

## **1.6. RESEARCH CONTRIBUTIONS**

This study aims to analyze the factors influencing the financial sustainability of Vietnamese commercial banks during the period from 2010 to 2023. The estimation techniques, including Feasible Generalized Least Squares (FGLS), Ordinary Least Squares (OLS), Fixed Effects Model (FEM), Random Effects Model (REM), and quantile regression analysis, are applied to a panel dataset collected from the financial statements of 27 Vietnamese commercial banks over the 2010–2023 period.

Building upon theoretical foundations and prior studies related to this topic, this research provides an in-depth analysis of the extent to which these factors impact the financial sustainability of Vietnamese commercial banks. The findings of this study will serve as the basis for discussions on potential solutions and policy implications aimed at enhancing the financial sustainability of commercial banks in Vietnam.

## **1.7. RESEARCH STRUCTURE**

### **Chapter 1: Introduction**

Chapter 1 provides a general introduction to the research, including the rationale for selecting the topic, research objectives, research questions, research scope and subjects, research methodology, contributions of the study, and the structure of the dissertation.

### **Chapter 2: Theoretical Foundation And Literature Review**

The content of chapter 2 aims to outline the theoretical framework on the financial sustainability of commercial banks and the factors influencing their financial sustainability. Additionally, the author reviews relevant previous studies related to the research topic, providing a foundation for developing the model for this thesis.

### **Chapter 3: Research Model and Methodology**

Building upon theoretical foundations and previous research, chapter 3 introduces an appropriate research model for the study. It specifies the variables included in the model, describes the dataset, and details the research methodology employed.

### **Chapter 4: Research Results and Discussion**

Based on the research model and methodology outlined in chapter 3, chapter 4 presents the results of descriptive statistical analysis for the variables in the model, correlation analysis, and panel data regression analysis. This chapter also identifies the direction and magnitude of the impact of independent variables on the dependent variable, thereby analyzing the relationship between the examined factors and the financial sustainability of Vietnamese commercial banks. Additionally, it addresses econometric issues such as heteroskedasticity, autocorrelation, and endogeneity.

### **Chapter 5: Conclusion and Implications**

Based on the results of the SGMM estimation and quantile regression analysis on the factors influencing the financial sustainability of Vietnamese commercial banks in chapter 4. Chapter 5 presents the key conclusions of the study. It then provides policy implications for Vietnamese commercial banks to enhance their financial sustainability. Additionally, chapter 5 discusses the study's limitations and suggests directions for future research.

## **SUMMARY OF CHAPTER 1**

Chapter 1 provided an overview of the topic “The factors influencing the financial sustainability Of Vietnamese Commercial Banks: A quantile regression analysis” by presenting the rationale for selecting the research topic, outlining the research objectives, clearly defining the research subjects and scope, describing the research methodology, and finally, detailing the structure of the dissertation.

## **CHAPTER 2: THEORETICAL FRAMEWORK AND LITERATURE REVIEW**

### **2.1. THEORETICAL FRAMEWORK**

#### ***2.1.1. Definition of financial sustainability in banking***

Although extensive research on financial sustainability in banking has been conducted internationally. However, there is still no universally agreed-upon definition for this concept. Previous studies approach financial sustainability from various perspectives, highlighting different dimensions of its meaning and implications. Such as:

Jo et al. (2015) associate financial sustainability in commercial banks with competitiveness. According to their view, a financially sustainable bank is one that improves continuously its performance and business operations, thereby achieving and maintaining the first position in the industry. Meanwhile, Berger and Bouwman (2009) emphasize the role of effective risk management, arguing that a financially sustainable bank must be capable of identifying, measuring, and managing risks comprehensively to reduce potential losses and ensure operational stability.

A bank is considered financially sustainable when it maintains sufficient capital reserves to absorb economic shocks and ensure continuous operations even in adverse conditions. In this regard, Demirguc-Kunt and Huizinga (2010) focus on the importance of a strong capital structure as a determinant of financial sustainability. Similarly, Nurhikmah and Rahim (2021) define financial sustainability as a measure of a bank's ability to maintain and enhance its financial performance, emphasizing the crucial role of equity capital growth in ensuring long-term stability.

Shaikh et al. (2022) examine the impact of Fintech on financial sustainability, suggesting that a sustainable bank must adapt and use financial technologies to

improve operational efficiency, enhance risk management, and meet evolving customer demands. Meanwhile, Ghenimi, Chaibi, and Omri (2021) explore the relationship between corporate governance and bank performance in developing economies. Their findings suggest that a financially sustainable bank must have a transparent and responsible governance structure, which enhances operational efficiency and profitability.

In summary, there is still no widely accepted definition of financial sustainability in banking. However, different perspectives help explain its various aspects. Financial sustainability can be understood through factors such as competitiveness, risk management, capital structure, operational efficiency, technological adaptation, and corporate governance. Each study provides valuable insights, emphasizing the need for a strong, stable, and efficient banking system that can withstand economic changes and market uncertainties. The diversity of research perspectives also highlights the complexity and multidimensional nature of this concept, requiring further in-depth study and analysis in the future.

### ***2.1.2. Measuring financial sustainability of commercial banks***

According to Ongore & Kusa (2013), the financial sustainability of banks plays a crucial role in the economic well-being of any country. It is often considered not only a reflection of a nation's financial and economic conditions but also a key component of its financial intermediation system. Financial sustainability is a complex and diverse measurement indicator, serving as a fundamental factor that is closely interconnected and mutually supportive with ESG index (Environmental, Social, and Governance index). Various authors have measured bank sustainability in different ways in their studies, such as:

Moechdi et al. (2016) assessed financial sustainability using three key indicators: profitability (ROA – return on assets), loan quality (NNPL – Net Non-

Performing Loan Ratio - NPLs net of credit risk provisions to total outstanding loans), and operational self-sufficiency (OSS - Total operating income to total operating expenses and write-off costs, expressed as a percentage). The study utilizes nine independent variables, categorized into two groups: internal variables and classification variables.

The study by Meher and Getaneh (2019) suggests that the variables representing financial sustainability include Return on Assets (ROA), Return on Equity (ROE), Bank Soundness (BS), and the Financial Stability Index (FSI). Meanwhile, the research by Sitnikova et al. (2020) measures financial sustainability using four indicators, ensured by the comprehensive presence of key factors, including asset quality, liability quality, liquidity quality, and profitability.

Zavadzka (2011) argued that financial sustainability management in banking should focus on core elements such as capital and resources, ensuring viability and resilience against risks. Unlike Zavadzka, Sheludko (2002) employed a multivariate approach, highlighting the importance of performance indicators in evaluating a bank's financial sustainability. Similarly, Kornyluk (2013) considered scale and operational efficiency over a defined period as the primary indicators for assessing financial sustainability in the banking sector.

Studies by Nurhikmah & Rahim (2021) and Hamad & Mikhilif (2023) utilized the CAMELS framework, a widely applied assessment tool in the banking industry, to analyze financial sustainability. This approach evaluates six critical components: Capital Adequacy, Asset Quality, Management Quality, Earnings, Liquidity, and Sensitivity to Market Risk. However, both studies primarily focused on four key indicators: capital, asset quality, earnings, and liquidity.

Two studies by Sanfa & Ida (2023) and Arifani (2020) employed the total financial income to financial expense ratio as a measure of financial sustainability. The Financial Sustainability Ratio (FSR) was used to assess a bank's ability to sustain operations and generate long-term profits based on financial performance. A financially sustainable bank must continuously seek expansion opportunities to enhance revenue generation. FSR helps evaluate a bank's potential for profitability growth, thereby ensuring long-term operational viability. As Widyaningsih (2012) stated, financial sustainability is a critical factor in determining a bank's ability to operate continuously in the future. Financial Sustainability Ratio (FSR) is calculated based on data from financial statements, particularly the income statement and the balance sheet. If  $FSR > 100\%$ , meaning total income exceeds total expenses within a given period, the bank is considered to be in a strong financial sustainability position.

Financial Sustainability Ratio (FSR) Formula

$$FSR = \frac{\text{Total finance income}}{\text{Total finance expenses}}$$

*In which:*

**Total Finance Income** represents the sum of all income generated from core financial intermediation activities, primarily including interest earned on loans and investments, along with income from trading activities, fees, and commissions related to financial services.

**Total Finance Expenses** represents the sum of all expenses incurred in raising and managing funds, including interest paid on deposits and borrowings, as well as costs associated with loan loss provisions and other financial operations.

In summary, in the context of intense competition and constant volatility in today's financial markets, maintaining profitability serves as evidence of a bank's adaptability, operational efficiency, and effective risk management. For this reason, the FSR model has been employed in this study to measure the financial sustainability

of commercial banks in Vietnam, particularly in the analysis of factors influencing financial sustainability in the Vietnamese banking sector. Commercial banks play a central role in the banking system, serving as the primary institutions for mobilizing capital, providing credit, and delivering core financial services within the economy. Therefore, financial assessment models and indicators developed for banks in general can be reasonably inferred and applied specifically to the context of commercial banks.

### ***2.1.3. Theories of factors affecting the financial sustainability of commercial banks***

The CAMELS model is a popular evaluating system used by banking regulators to assess the financial condition and operational performance of banks. This model includes six key components: Capital Adequacy (C), Asset Quality (A), Management Quality (M), Earnings (E), Liquidity (L), and Sensitivity to Market Risk (S). This framework is essential for identifying financial vulnerabilities, ensuring regulatory compliance, and enhancing the long-term financial sustainability of banks. This model helps identify the strengths and weaknesses of banks, thereby providing recommendations to improve operations and enhance financial sustainability.

Agency Theory examines the relationship between the principal (owner), who delegates authority to an agent (representative) to act on their behalf, and the conflicts of interest that arise due to the separation of ownership and control. This delegation often results in agency conflicts, arising from the separation of ownership and control, leading to potential misalignment of interests. The costs associated with these conflicts, known as agency costs, include monitoring costs, bonding costs, and residual loss. To mitigate these costs, governance mechanisms such as performance-based compensation and independent board oversight are implemented to align managerial incentives with shareholder interests. In the banking sector, Agency

Theory is useful to explain relevant the risk-taking behavior by bank executives, especially in situations of conflicting interests between shareholders and management. For example, the study by John, Saunders, and Senbet (2000) indicates that higher agency costs, arising from conflicts between shareholders and management, may lead banks to take on greater risks, thereby threatening the safety and stability of banking operations.

## **2.2. RELATED STUDIES**

### ***2.2.1. Foreign research***

Meher and Getaneh (2019) demonstrated that the financial sustainability of banks is influenced by a range of financial and risk-related factors. These include absolute liquidity risk, overall liquidity risk, asset quality, net interest rate risk, credit risk, solvency risk, the growth rate of net income, and the capital adequacy ratio. Each of these factors affects a bank's ability to maintain long-term operational viability, though their impact may vary across different banking institutions and regulatory environments.

Arifani (2020) examined the impact of Non-Performing Loans, Loan-to-Deposit Ratio, Operating Expenses to Operating Income, and Net Interest Margin on the Financial Sustainability Ratio of Islamic commercial banks in Indonesia. However, this study had some limitations, as it only considered four internal banking performance indicators and focused on a limited sample of 13 Islamic commercial banks over three years. To address these limitations, Sanfa and Ida (2023) expanded the study by including ROA and CAR as independent variables and extended the research scope to conventional commercial banks in Indonesia from 2019 to 2021.

Kiptum (2023) explored the relationship between economic diversification and banks' financial sustainability, particularly examining the moderating role of debt structure. The study sought to answer key questions such as whether expanding into

multiple business areas strengthens financial stability and how debt structure influences this relationship.

Ahamed (2017) provided compelling evidence on the benefits of diversification in Indian banks. By expanding beyond traditional lending activities into non-traditional areas such as insurance, securities investment, and asset management, these banks significantly increased their non-interest income. This highlights that diversification is not only a defensive strategy for risk mitigation but also a powerful driver of growth, enhancing revenue streams and boosting banks' competitiveness. At the same time, it enhances financial sustainability.

Ali and Puah (2018) examined the negative impact of bank size on stability, which was estimated using the Z-score model. Additionally, funding risk exhibited a positive relationship, while inflation and GDP growth had a negative effect on the financial stability of banks in Pakistan. Financial sustainability is a broader concept that encompasses financial stability. A financially sustainable bank must be able to maintain stable operations in the long term, including resilience to shocks and risks.

Ishaq et al. (2016) applied the CAMEL approach to assess the performance of commercial banks in Pakistan. Using a linear regression model, the authors examined EPS (Earning Per Share) as the dependent variable, while the independent variables consisted of key CAMEL framework components such as capital adequacy, asset quality, management efficiency, earnings, and liquidity. The results showed that financial ratios based on the CAMEL framework had a significant impact on banks' performance. In particular, Return on Assets and Return on Equity positively influenced banks' financial performance, which in turn contributed to financial sustainability.

### ***2.2.2. Domestic research***

Nguyen Thi Thuy (2023) examined the factors influencing the Financial Self-Sustainability (FSS) of Micro - Finance institutions (MFIs) in Vietnam, using data from 24 MFIs from 2016 to 2021. The findings indicated that capital adequacy ratio and income positively affected FSS, while credit growth, non-performing loan ratio, and economic growth had negative impacts. The study provided important recommendations for enhancing the financial self-sustainability of MFIs and helping managers better control these impacts.

Nguyen et al. (2012) examined the effect of business model selection on bank stability in ASEAN countries. Using System Generalized Method of Moments (SGMM) and other econometric techniques on a sample of 99 commercial banks, the study found that diversification into non-interest and fee-based activities had a significant negative impact on bank stability. However, the study was conducted in 2012, and economic and financial market conditions have likely changed since then, affecting the relevance of its findings.

Tran Thi Hoang Yen (2016) employed a multiple linear regression model to examine the relationship between corporate social responsibility (CSR) and banks' financial performance, using ROA and ROE as performance indicators. The study adopted the stakeholder theory to analyze CSR practices according to international standards. However, the research primarily focused on the financial performance impact of CSR rather than providing a comprehensive assessment of financial sustainability.

Similarly, Nguyen Viet Hung (2008) used both qualitative and quantitative approaches, including Stochastic Frontier Analysis (SFA), Data Envelopment Analysis (DEA) and a Tobit model to assess the efficiency of Vietnamese commercial banks. However, the study was limited to efficiency evaluation and did not provide a holistic view of financial sustainability.

Research on the financial sustainability of the commercial banking system in Vietnam remains relatively limited and lacks depth in clarifying different levels and models of sustainability. Most existing studies focus on evaluating operational efficiency, while research on the actual state of financial sustainability is still scarce. This indicates a significant gap in the study of financial sustainability in Vietnamese commercial banks, particularly in the context of international economic integration and the development of Fintech.

### ***2.2.3. Factors affecting the financial sustainability of commercial banks***

**Bank Size (lnSIZE):** Moechdi et al (2016) emphasize the importance of assets as a key indicator of bank size in financial sustainability. However, the relationship between size and financial sustainability is not always linear or positive. While an increase in size offers various advantages, it also introduces new challenges, particularly in risk management. Larger banks, with abundant resources and extensive market access, tend to engage in more complex and higher-risk financial activities compared to smaller banks, as analyzed by Daoud and Kammoun (2020).

**Return on Assets (ROA):** Profitability in relation to total assets reflects financial sustainability in banks, indicating stable income growth over the years (Meher & Getaneh (2019); Moechdi et al (2016)). A high and stable ROA suggests that a bank operates efficiently, generates strong profits, and has a greater capacity to withstand risks. Conversely, a low or negative ROA threatens the bank's viability and long-term sustainability (Shakhrun et al. (2019); Sanfa & Ida (2023)).

**Capital Adequacy Ratio (CAR):** The study of Munandar and Aravik (2022) illustrate a significant positive correlation between CAR and financial sustainability. A high CAR reflects a bank's ability to maintain strong core capital, ensuring compliance with regulatory standards while fortifying its stability against financial fluctuations. Additionally, domestic research by Nguyen Thi Thuy (2023) on microfinance institutions also affirms the positive relationship between CAR and financial sustainability.

**Non-performing loans (NPL):** NPL play a crucial role in assessing a bank's financial stability. According to Rajan and Dhal (2003), this indicator not only reflects the quality of a bank's asset portfolio but also poses significant challenges in risk management and directly impacts the efficient allocation of resources for business operations. The inability of borrowers to repay their loans presents a considerable threat to banks if not addressed promptly and effectively. As highlighted by Mabruroh (2004), banks may suffer severe financial losses if non-performing loans are not managed properly. However, studies by Santoso (2017) and Arifani (2020) argue that non-performing loans (NPLs) do not significantly impact financial sustainability. Their findings, based on multiple linear regression analysis, indicate an insignificant probability value of 0.395 ( $> 0.025$ ), suggesting no statistically meaningful relationship between NPLs and financial sustainability.

**Liquidity Ratio (LIQ):** LIQ plays a pivotal role in assessing the financial sustainability of commercial banks for two key reasons. First, ample liquidity enables banks to meet new credit demands without the need to recall existing loans. Second, it ensures that banks have sufficient funds to fulfill withdrawal requests from depositors at any time. The study by Rajhi and Hassairi (2013) demonstrated a positive correlation between liquidity and a bank's financial sustainability: the higher the liquidity, the greater the bank's ability to maintain long-term operational stability. However, it is important to note that holding excessive cash reserves can negatively impact a bank's profitability. Therefore, achieving an optimal balance between liquidity and profitability remains a critical challenge for commercial banks.

**Operating Income (lnINC):** The author utilized the Financial Sustainability Ratio (FSR) formula to determine that operating income influences the financial sustainability of commercial banks. This is because it affects all aspects of a bank's operations, including profitability, risk management, investment in development, liquidity, and market reputation. According to Nguyen Thi Thuy (2023), who examined factors influencing the financial self-sustainability (FSS) of microfinance

institutions (MFIs), operating income has a positive correlation with financial sustainability.

**Loan-to-Deposit Ratio (LTD):** Hesse and Čihák (2007) highlighted an inverse relationship between the loan-to-deposit ratio and a bank's financial stability. Specifically, banks with a high LDR, meaning that a significant portion of their assets is allocated to lending activities, tend to face an increased risk of non-performing loans. This leads to higher credit risk, weakening profitability and negatively impacting long-term financial sustainability. However, Vong and Chan (2009) offered a different perspective by emphasizing growth potential. Their research suggested that effective credit portfolio expansion, reflected in healthy loan asset growth, can generate substantial income for banks. This implies that prudent and efficient loan portfolio management is key to balancing risk and return, thereby strengthening financial sustainability. In other words, credit growth is not inherently bad; the key issue lies in the quality of that growth.

**Board Independence ratio (BI):** Although no existing research has specifically examined the impact of the ratio of independent board members on the financial sustainability of commercial banks, this study aims to clarify this relationship, as it directly affects corporate governance quality. Independent board members, with their objectivity and lack of ties to executive management or major shareholders, play a crucial role in oversight, control, and impartial decision-making. This enhances operational efficiency and mitigates risks for banks. Several major financial scandals worldwide have resulted from weak governance and a lack of independent oversight from the board of directors (BOD). For example, the collapses of Enron and WorldCom highlighted the severe consequences of boards failing to fulfill their supervisory roles effectively. While these crises were not solely due to the absence of independent members, the lack of effective BOD oversight was one of the key factors leading to financial turmoil. These cases underscore the importance

of a strong and independent board in risk prevention and ensuring financial sustainability.

**Economic Growth (GDP):** Rupeika-Apoga et al. (2018) demonstrated the significant positive impact of GDP growth on the long-term stability of the banking sector. Their findings highlight the crucial role of macroeconomic development in strengthening the resilience and overall robustness of the financial system, including commercial banks. A growing economy creates a favorable business environment, minimizes credit risk, and promotes financial sustainability in the banking industry. However, Meher and Getaneh (2019) offer a different perspective, arguing that financial sustainability, when considered independently of systemic risk, is not significantly influenced by macroeconomic factors, including GDP. This suggests that, beyond macroeconomic conditions, internal factors specific to each bank such as governance quality, operational efficiency, and risk management are also critical in ensuring financial stability and sustainability.

**Inflation rate (INF):** INF has a significant impact on the economy, particularly on the banking sector. High inflation leads to currency depreciation, reducing the real value of the debts and assets held by banks, thereby increasing credit risk as customers' debt repayment capacity deteriorates. While Tu and Oanh (2021) found a positive impact of inflation on banking stability, contributing to financial sustainability, numerous other studies have highlighted its negative effects. Specifically, inflation can drive up interest rates, which, although increasing banks' interest income, simultaneously restricts customers' access to credit, thereby affecting credit growth. The research findings of Lan (2021) and Rahim et al. (2012) both indicate a negative impact of inflation on financial sustainability of commercial banks. Nkusu (2011) argued that the effect of inflation on borrowers' repayment ability is twofold: on one hand, high inflation reduces the real value of debt, making it easier for borrowers to repay; on the other hand, if inflation erodes real income

(e.g., when wages fail to keep up with inflation), borrowers' repayment capacity may also decline.

**Table 2. 1 Summary of domestic and foreign research**

| <b>Author(s)</b>       | <b>Methodology</b>                  | <b>Independent Variables</b>                                                                                       | <b>Research Results</b>                       |
|------------------------|-------------------------------------|--------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| Meher & Getaneh (2019) | Quantitative analysis               | Liquidity risk<br>Asset quality<br>Net interest margin<br>Credit risk<br>Solvency risk<br>Net income growth<br>CAR | (-)<br>(+)<br>(+)<br>(-)<br>(-)<br>(+)<br>(-) |
| Arifani (2020)         | Panel data (13 banks, 3 years)      | NPL<br>LDR<br>OEOI<br>NIM                                                                                          | (-)<br>(+)<br>(+)<br>(+)                      |
| Sanfa & Ida (2023)     | Extended panel regression           | ROA<br>CAR<br>NPL<br>LDR<br>OEOI<br>NIM                                                                            | (+)<br>(+)<br>(-)<br>(+)<br>(+)<br>(-)        |
| Kiptum (2023)          | Regression with moderation analysis | Economic diversification<br>Debt structure                                                                         | (+)<br>(+)                                    |
| Ahamed (2017)          | Empirical analysis                  | Income sources<br>Diversification (non-interest income)                                                            | (+)<br>(+)                                    |
| Ali & Puah (2018)      | Z-score model                       | Bank size<br>Funding risk<br>Macroeconomic indicators (inflation, GDP)                                             | (-)<br>(-)<br>(+)<br>(-)                      |
| Ishaq et al. (2016)    | CAMEL model + linear regression     | CAMEL indicators<br>EPS<br>ROA<br>ROE                                                                              | (-)<br>(+)<br>(+)                             |

|                                                                           |                                                                |                                                                                                   | (+)                             |
|---------------------------------------------------------------------------|----------------------------------------------------------------|---------------------------------------------------------------------------------------------------|---------------------------------|
| Nguyen Thi Thuy (2023)                                                    | Panel regression                                               | CAR<br>Income<br>Credit growth<br>NPL<br>GDP growth                                               | (+)<br>(+)<br>(-)<br>(-)<br>(+) |
| Tran Thi Hoang Yen (2016)                                                 | Multiple regression                                            | CSR<br>ROA<br>ROE                                                                                 | (+)<br>(+)<br>(+)               |
| Nguyen Viet Hung (2008)                                                   | SFA, DEA, Tobit                                                | Operational efficiency                                                                            | (+)                             |
| <i>Factors affecting the financial sustainability of commercial banks</i> |                                                                |                                                                                                   |                                 |
| Factor                                                                    | Impact on Financial Sustainability                             | Key References/Notes<br>Factor                                                                    |                                 |
| Bank Size (lnSIZE)                                                        | (+)<br>(-)                                                     | Moechdi et al. (2016);<br>Daoud & Kammoun (2020)                                                  |                                 |
| Return on Assets (ROA)                                                    | (+)<br>(+)<br>(+)<br>(+)                                       | Meher & Getaneh (2019);<br>Moechdi et al. (2016);<br>Shakhrun et al. (2019)<br>Sanfa & Ida (2023) |                                 |
| Capital Adequacy Ratio (CAR)                                              | (+)<br>(+)                                                     | Munandar & Aravik (2022);<br>Nguyen Thi Thuy (2023)                                               |                                 |
| Non-performing Loans (NPL)                                                | (-)<br>(-)<br>no significant impact.<br>no significant impact. | Rajan & Dhal (2003);<br>Mabruroh (2004);<br>Santoso (2017);<br><br>Arifani (2020)                 |                                 |
| Liquidity Ratio (LIQ)                                                     | (+)                                                            | Rajhi & Hassairi (2013)                                                                           |                                 |
| Operating Income (lnINC)                                                  | (+)                                                            | Nguyen Thi Thuy (2023)                                                                            |                                 |

|                               |                          |                                                                          |
|-------------------------------|--------------------------|--------------------------------------------------------------------------|
| Loan-to-Deposit Ratio (LTD)   | (-)<br>(+)               | Hesse & Čihák (2007);<br>Vong & Chan (2009)                              |
| Board Independence Ratio (BI) | x                        | No direct studies                                                        |
| Economic Growth (GDP)         | (+)<br>(+)               | Rupeika-Apoga et al. (2018);<br>Meher & Getaneh (2019)                   |
| Inflation Rate (INF)          | (+)<br>(-)<br>(-)<br>(-) | Tu & Oanh (2021);<br>Lan (2021);<br>Rahim et al. (2012);<br>Nkusu (2011) |

*Source: Compiled by the author*

#### **2.2.4. Research gap**

Although financial sustainability in banking has been widely studied globally, research on Vietnamese commercial banks remains limited and fragmented. Existing studies primarily focus on bank performance and financial stability, lacking a comprehensive examination of long-term sustainability and its key determinants. While international research highlights factors such as capital adequacy, income diversification, and governance, empirical studies in Vietnam have yet to systematically analyze their impact. Furthermore, previous research often relies on traditional econometric models like OLS or fixed effects, which may not fully capture heterogeneous effects and dynamic relationships in financial sustainability. Additionally, corporate governance, particularly the role of independent board members, remains underexplored, despite its crucial role in ensuring financial resilience. This study aims to fill these gaps by employing advanced econometric techniques (SGMM and Quantile Regression) to assess the multidimensional impact of both internal and external factors on financial sustainability. The findings provide empirical evidence and policy recommendations to help banks strengthen risk management, governance, and long-term resilience in the face of economic fluctuations and increasing fintech development.

## **SUMMARY OF CHAPTER 2**

Chapter 2 presented the theoretical foundations, including core concepts and key theoretical models related to the financial sustainability of banks. Additionally, this chapter provided an overview of empirical studies, both domestic and international, examining the factors influencing financial sustainability in banking. The analysis and synthesis of these research findings play a crucial role in identifying key variables and constructing the analytical framework for this study. Based on these foundations, this chapter has established the groundwork for developing the research model and selecting the appropriate research methodology, which will be detailed in chapter 3.

## **CHAPTER 3: RESEARCH MODEL AND METHODOLOGY**

### **3.1. DATA SOURCES**

The data collected by the author is classified into two groups:

**Microeconomic Data:** This data is obtained from audited and officially published financial statements, income statements, and annual reports of 27 Vietnamese commercial banks from 2010 to 2023. The sources include websites such as Vietstock and the official websites of the commercial banks examined in this study.

**Macroeconomic Data:** This includes indicators such as GDP growth and inflation rates, collected from reputable sources such as the State Bank of Vietnam (SBV), the World Bank, and the Asian Development Bank (ADB), which provide reliable data specifically for ASEAN countries.

### **3.2. RESEARCH METHOD**

The author employed two research methods to measure the factors affecting the financial sustainability of commercial banks:

First, the author utilized a qualitative research approach to review, synthesize, and analyze previous studies conducted both domestically and internationally. Through reviewing prior research, the author was able to propose a suitable research model for further in-depth investigation.

Next, a quantitative research method was applied using Stata 17 statistical software to analyze and summarize the characteristics of the dataset, including the mean, maximum, minimum, and standard deviation of both dependent and independent variables. Subsequently, the author conducted various tests, such as correlation matrix analysis, heteroscedasticity test, multicollinearity test, autocorrelation test, and model selection test. In cases where endogeneity issues were

detected, the System Generalized Method of Moments (SGMM) proposed by Blundell and Bond (1998) was employed to address these limitations. Finally, the author performed quantile regression using the `qreg` command as an alternative to OLS regression, allowing for a more comprehensive assessment of the impact of factors across different quantiles rather than relying solely on the mean, particularly in cases where the data distribution was skewed.

### ***3.2.1. Descriptive statistics***

Descriptive statistics are used to provide an overview of the variables in the research model, including mean, observation, maximum, standard deviation, and minimum. These indicators help evaluate the distribution of each variable, check for anomalies in the dataset, and identify key characteristics of the data. For instance, if a variable has a high standard deviation, it indicates significant fluctuations, which may affect the stability of the regression model. Conversely, if there is a substantial difference between the mean and median values, it suggests a skewed distribution, requiring careful consideration during analysis.

Additionally, descriptive statistics help detect the presence of outliers, enabling appropriate data processing decisions, such as removing invalid observations or applying missing data handling techniques to ensure the accuracy of research results.

Overall, descriptive statistics not only provide a general overview of the dataset but also play a crucial role in assessing data quality, thereby enhancing the reliability of subsequent analyses.

### ***3.2.2. Correlation***

The research team employs the `corr` command to determine the linear relationship between two or more variables. In other words, the correlation coefficient indicates the strength of the relationship between multiple variables within the model. The correlation coefficient ranges from -1.0 to 1.0, and if the calculated result exceeds this range, it suggests an error in the correlation measurement:

- A negative correlation coefficient implies an inverse relationship (negative correlation), with a perfect inverse correlation occurring when the value is -1.
- A positive correlation coefficient indicates a direct relationship (positive correlation), with a perfect direct correlation occurring when the value is 1.

If the correlation coefficient between independent variables exceeds 0.8, it suggests a strong linear relationship, which may lead to multicollinearity issues, potentially affecting the research outcomes.

### ***3.2.3. Panel data regression, testing for model violations and addressing the violations of model***

#### ***3.2.3.1. Panel data regression***

#### **Pooled Ordinary Least Squares (Pooled OLS)**

Pooled Ordinary Least Squares (Pooled OLS) applies the standard OLS method to panel data, treating it as a single cross-sectional sample. It estimates one regression equation for all individuals and time periods, ignoring individual-specific and time-specific effects.

This method is appropriate when unobserved heterogeneity is negligible. However, if individual characteristics influence the dependent variable, Pooled OLS may lead to biased estimates due to omitted variable bias. Researchers often use statistical tests, such as the Breusch-Pagan LM test or the F-test for fixed effects, to determine whether more advanced models like Fixed Effects (FE) or Random Effects (RE) are needed.

While Pooled OLS is computationally simple and serves as a useful baseline model, its assumption of homogeneity may not hold in practice, potentially leading to inefficiencies if individual differences are significant.

### **Fixed Effects Model (FEM)**

The Fixed Effects Model (FEM) is a widely used regression technique in panel data analysis that accounts for unobserved individual characteristics unique traits of each entity that remain constant over time. These characteristics, such as corporate culture, geographic location, or education level, may influence the dependent variable but are difficult to measure explicitly. If these effects are ignored, omitted variable bias can arise, leading to unreliable estimates.

According to Pham Tuyet Trinh (2016), FEM is the preferred method when panel data entities exhibit heterogeneity. This model assumes that the regression coefficients of independent variables remain the same across individuals, but each entity has its own distinct intercept, capturing intrinsic differences. By allowing intercepts to vary, FEM controls for unobservable factors that could bias the estimates.

A key advantage of FEM is its ability to mitigate endogeneity concerns caused by unobserved time-invariant factors. However, a major limitation is that FEM does not estimate the effect of time-invariant variables (e.g., industry classification, geographic location) because these factors are absorbed into the individual-specific intercepts.

### **Random Effects Model (REM)**

The Random Effects Model (REM) offers an alternative approach to handling individual effects in panel data. Like FEM, it accounts for repeated observations over time for the same entities, but it differs in its treatment of unobserved heterogeneity. Instead of assuming entity-specific effects are fixed constants, REM treats them as random variables drawn from a common distribution.

A crucial assumption in REM is that individual effects are uncorrelated with the independent variables in the model. This implies that entity-specific characteristics do not systematically influence the relationship between explanatory

variables and the dependent variable. If this assumption holds, REM is more efficient than FEM because it retains degrees of freedom and allows for the estimation of time-invariant variables. However, if individual effects are correlated with independent variables, REM produces biased estimates, making FEM the more reliable choice.

### **3.2.3.2. *Testing for model violations***

#### **Model selection tests**

To determine the most appropriate regression model for panel data analysis, the author conducts a series of statistical tests, including the F-test, Hausman test, and Breusch-Pagan test. These tests facilitate model selection by comparing different approaches:

- F-test: Determines whether the Fixed Effects Model (FEM) is preferable to Ordinary Least Squares (OLS) by testing for the presence of entity-specific effects.
- Hausman test: Compares FEM and the Random Effects Model (REM) to assess whether individual effects are correlated with the explanatory variables. A significant result favors FEM, while an insignificant result supports REM.
- Breusch-Pagan test: Evaluates whether REM is preferable to OLS by testing for the presence of random effects. If the null hypothesis of no random effects is rejected, REM is a better fit than OLS.

By systematically applying these tests, the author ensures that the chosen model accurately captures the data structure and accounts for individual effects.

#### **Variance Inflation Factor (VIF) test**

Multicollinearity, a common issue in regression analysis, occurs when independent variables are highly correlated, leading to unreliable coefficient estimates and inflated standard errors. To detect multicollinearity, the author applies the Variance Inflation Factor (VIF) test.

- A VIF greater than 2 signals potential multicollinearity, while a VIF exceeding 8 strongly indicates its presence.
- High VIF values can distort coefficient estimates, widen confidence intervals, and reduce the interpretability of the R-squared ( $R^2$ ) value.

To address multicollinearity, the author considers removing variables with excessively high VIF values. However, variable elimination is not based solely on statistical criteria; economic theory and practical significance play a crucial role in determining which variables remain in the model.

### **Heteroscedasticity test**

A key assumption of regression models is homoscedasticity, meaning that the variance of the error terms remains constant across all levels of the independent variables. Heteroscedasticity arises when this assumption is violated, leading to inefficient estimates and misleading statistical inferences. To detect heteroscedasticity, the author applies:

- The Wald test for the Fixed Effects Model (FEM).
- The Breusch-Pagan test for the Random Effects Model (REM).

If heteroscedasticity is detected, robust standard errors or alternative estimation techniques may be used to improve the reliability of the regression results.

### **Autocorrelation test**

Autocorrelation occurs when the error terms in a regression model are correlated across time, violating the assumption of independence. This issue is particularly relevant in panel data and time-series models, where past errors can influence future errors.

To test for autocorrelation, the author employs the Wooldridge test, a widely accepted method for detecting serial correlation in panel data models. Identifying and addressing autocorrelation is essential for ensuring accurate hypothesis testing and reliable coefficient estimates.

### **Endogeneity test**

Endogeneity arises when an independent variable ( $X$ ) is correlated with the error term, leading to biased and inconsistent estimates. Common causes include omitted variable bias, measurement errors, and simultaneity.

To assess endogeneity, the author conducts formal statistical tests. If endogeneity is detected, the System Generalized Method of Moments (SGMM) approach is applied to mitigate its effects and produce unbiased parameter estimates. This technique is particularly useful in dynamic panel data models where lagged dependent variables or endogenous regressors are present.

By systematically performing these diagnostic tests, the author ensures the robustness, accuracy, and reliability of the estimated regression models, providing strong empirical evidence for the study.

#### ***3.2.3.3. Addressing the violations of model***

### **Feasible Generalized Least Squares (FGLS)**

Feasible Generalized Least Squares (FGLS) extends the Ordinary Least Squares (OLS) method to address common econometric issues, particularly heteroscedasticity (non-constant variance of error terms) and autocorrelation (correlation of errors across observations). When key OLS assumptions are violated such as when the variance of the error terms is not constant or when error terms exhibit correlation over time OLS estimators become inefficient. This inefficiency results in biased standard errors, leading to unreliable hypothesis testing and confidence intervals.

FGLS corrects these shortcomings by estimating the error variance-covariance matrix and applying appropriate weighting, thereby improving the precision of coefficient estimates. Compared to OLS, FGLS produces more efficient estimators with reduced variance, making it particularly useful in panel data and time-series analysis. However, despite its advantages, FGLS has limitations. Notably, it does not directly address endogeneity, a critical econometric issue that arises when an

independent variable is correlated with the error term. In such cases, more advanced methods, such as the System Generalized Method of Moments (SGMM), are preferred.

### **System Generalized Method of Moments (SGMM)**

System Generalized Method of Moments (System GMM) is an advanced estimation technique designed to address endogeneity in dynamic panel data models. It improves upon Difference GMM by combining equations in both first differences and levels, allowing for more efficient use of available instruments.

System GMM is particularly useful when explanatory variables are persistent over time or when endogeneity arises due to omitted variables, simultaneity, or measurement errors. By employing valid lagged instruments, it provides consistent and reliable estimates without requiring strong distributional assumptions.

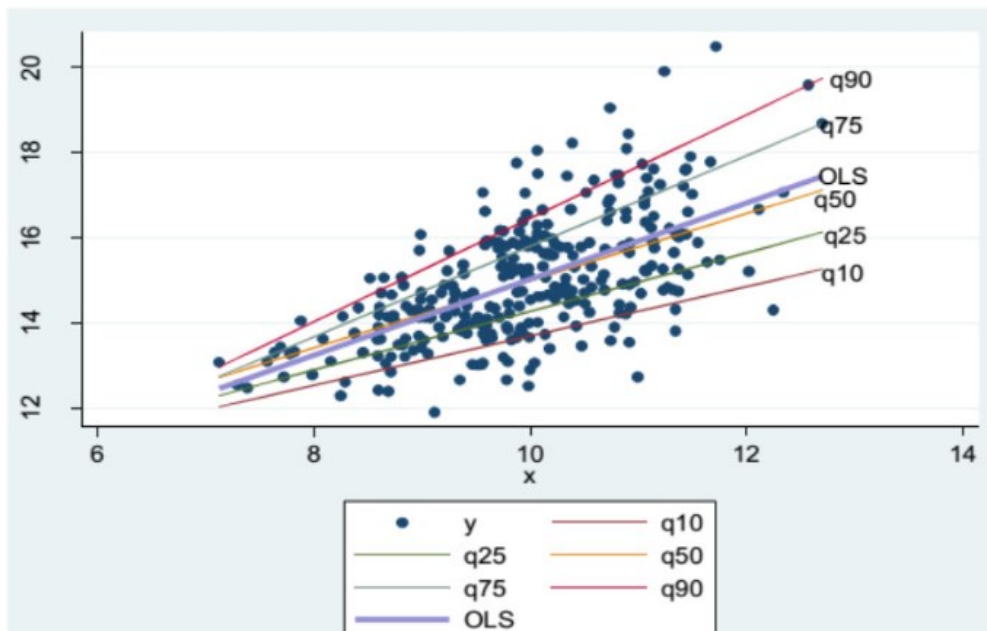
Thanks to its flexibility and robustness, System GMM is widely applied in economics and finance, especially where lagged dependent variables and endogenous regressors are present.

### **Quantile Regression (QR)**

Quantile Regression (QR), originally developed by Koenker and Bassett (1982), is employed in this study as an advanced econometric technique to explore the relationship between internal and external determinants and the financial sustainability of commercial banks across different levels of financial sustainability. Unlike the conventional Ordinary Least Squares (OLS) method, which estimates the average (conditional mean) effect of explanatory variables on the dependent variable, Quantile Regression allows the estimation of effects at various points (quantiles) in the distribution of the dependent variable. This provides a more nuanced and comprehensive understanding of how covariates influence the outcome variable across its entire distribution, rather than just at the center.

In Quantile Regression, instead of minimizing the sum of squared residuals as in OLS, the method minimizes the weighted absolute deviations, which varies by quantile. This approach is particularly effective in addressing issues of heteroskedasticity, outliers, and non-normality, which are common in financial data. Moreover, Quantile Regression enables researchers to detect heterogeneous effects of predictors—for instance, whether a given factor has a stronger influence on banks with low financial sustainability (e.g., lower quantiles) compared to more stable banks (e.g., upper quantiles).

Given the characteristics of the banking sector, where financial sustainability levels vary significantly across institutions, Quantile Regression is considered more suitable than OLS. The use of Quantile Regression in this study allows for the identification of asymmetric effects, which is particularly important for policy-making and strategic management. For example, a variable such as capital adequacy may exert a stronger stabilizing effect in underperforming banks, while its marginal effect may diminish in more stable institutions.



**Figure 3. 1. Graphical Representation of Quantile Regression Results of Y on X**

*Source: Compiled by the author from Internet*

### 3.3. RESEARCH MODEL

Based on previous studies by various authors on the factors affecting the financial sustainability of banks over different periods, the author has collected and selected a group of micro-level variables (within the bank) and macro-level variables (outside the bank) to measure the extent of their impact on the financial sustainability of commercial banks. The research model is constructed as follows:

$$FSR_{i,t} = \beta_0 + \beta_1 SIZE_{i,t} + \beta_2 ROA_{i,t} + \beta_3 CAR_{i,t} + \beta_4 NPL_{i,t} + \beta_5 LIQ_{i,t} + \beta_6 LnINC_{i,t} + \beta_7 LTD_{i,t} + \beta_8 BI_{i,t} + \beta_9 GDP_t + \beta_{10} INF_t + \mu_{i,t}.$$

*In which:*

- **FSR<sub>i,t</sub>**: The dependent variable representing the relationship between influencing factors and the financial sustainability of commercial banks.
- **SIZE<sub>i,t</sub>**: The size of bank *i* in year *t*.
- **ROA<sub>i,t</sub>**: Return on assets of bank *i* in year *t*.
- **CAR<sub>i,t</sub>**: Capital adequacy ratio of bank *i* in year *t*.
- **NPL<sub>i,t</sub>**: Non-performing loan of bank *i* at time *t*.
- **LIQ<sub>i,t</sub>**: Liquidity ratio of bank *i* at time *t*.
- **INC<sub>i,t</sub>**: Operating income of bank *i* at time *t*.
- **LTD<sub>i,t</sub>**: Loan-to-deposit ratio of bank *i* at time *t*.
- **BI<sub>i,t</sub>**: Independent board members ratio in bank *i* at time *t*.
- **GDP<sub>t</sub>**: Economic growth in year *t*.
- **INF<sub>t</sub>**: Inflation rate in year *t*.

### 3.4. RESEARCH VARIABLES AND HYPOTHESES

#### 3.4.1. *Dependent variables*

To assess the financial sustainability of commercial bank in Vietnam, this study uses the Financial Sustainability Ratio (FSR), similar to the approach used by Sanfa and Ida (2023) and Arifani (2020). FSR reflects a bank's profitability and long-

term operational viability based on financial performance, indicating whether the bank can expand its operations and increase its revenue. Study of Widyaningsih (2012) show that financial sustainability is a crucial factor for the continuous existence and future development of banks.

FSR is calculated as the ratio of total financial income to total financial expenses, extracted from the income statement and balance sheet:

$$\text{FSR} = \frac{\text{Total finance income}}{\text{Total finance expenses}}$$

If FSR exceeds 100% (indicating total income surpasses total expenses), the bank is in a financially sustainable position. In a highly competitive and volatile financial market, maintaining profitability and achieving a high FSR reflect a bank's adaptability, operational efficiency, and effective risk management. Therefore, FSR is selected as the financial sustainability indicator for commercial banks in this study.

#### ***3.4.2. Independent variables***

**Bank Size (lnSIZE):** is typically measured by total assets. According to the study by Moechdi et al. (2016), bank size has a positive impact on financial sustainability. However, the relationship between size and financial sustainability is not always linear and positive. As analyzed in the research by Daoud and Kammoun (2020), there is an inverse relationship between bank size and financial sustainability. Most theories suggest that larger banks benefit from market power and investment opportunities, leading to greater financial stability. However, in the context of Vietnam, this study anticipates the opposite effect. Specifically, large banks in Vietnam often engage in high-risk lending, which may reduce their financial sustainability. Therefore, author expects bank size to have a negative relationship with the financial sustainability of Vietnamese commercial banks.

$$\text{LNSIZE} = \text{Natural log of total assets}$$

**Hypothesis H1:** Bank size (lnSize) has a *negative impact* on the financial sustainability of Vietnamese commercial banks.

**Return on Assets (ROA):** is an indicator used to evaluate a bank's profitability from its total assets. This ratio is measured by dividing net profit after tax by total assets. A positive ROA indicates that the bank is generating profits, and a higher ROA suggests that the bank is utilizing its assets more efficiently to generate income. According to the studies by Meher & Getaneh (2019) and Moechdi et al. (2016), ROA positively affects to the financial sustainability of banks. Therefore, author expects that ROA has a positive relationship with the financial sustainability of banks. The formula for ROA is as follows:

$$\text{ROA} = \frac{\text{Net Profit After Tax}}{\text{Total assets}}$$

**Hypothesis H2:** Return on Assets (ROA) has a *positive impact* on the financial sustainability of Vietnamese commercial banks.

**Capital Adequacy Ratio (CAR):** According to Basel III standards, banks are required to strengthen their capital to mitigate risks and enhance the stability of the financial system. Studies by Fajari & Sunarto (2017) and Ayuningtyas et al. (2018) shows that a higher CAR contributes to financial stability and builds investor and market confidence. This study assumes that maintaining a high level of capital adequacy helps banks remain resilient against adverse financial shocks. Additionally, the domestic study by Thuy.T.T (2023) on microfinance institutions also found a positive relationship between CAR and financial sustainability. Therefore, author expects that CAR has a positive relationship with financial sustainability. The formula for CAR is as follows:

$$\text{CAR} = \frac{\text{Equity capital}}{\text{Risk – weighted assets}}$$

**Hypothesis H3:** Capital Adequacy Ratio (CAR) has a *positive impact* on the financial sustainability of Vietnamese commercial banks.

**Non-Performing Loan (NPL):** NPL plays a crucial role in assessing a bank's financial stability. According to Rajan & Dhal (2003), this indicator not only reflects the quality of a bank's asset portfolio but also presents significant challenges in risk management and directly impacts the efficient allocation of resources for business operations. Therefore, this study expects that NPL has a negative relationship with financial sustainability. The formula for NPL is as follows:

$$\text{NPL} = \frac{\text{Group 3 loans} + \text{Group 4 loans} + \text{Group 5 loans}}{\text{Total outstanding loans}}$$

**Hypothesis H4:** The Non-Performing Loan Ratio (NPL) has a *negative impact* on the financial sustainability of Vietnamese commercial banks.

**Liquidity Ratio (LIQ):** Rajhi and Hassairi (2013) demonstrated a positive relationship between liquidity and a bank's financial sustainability: the higher the liquidity, the greater the bank's ability to maintain stable operations. Therefore, author expects that liquidity has a positive relationship with financial sustainability. The formula for LIQ is as follows:

$$\text{LIQ} = \frac{\text{Current assets}}{\text{Total assets}}$$

**Hypothesis H5:** The Liquidity Ratio (LIQ) has a *positive impact* on the financial sustainability of Vietnamese commercial banks.

**Operating Income (lnINC):** Higher operating income enables banks to generate greater profits, strengthen equity capital, and enhance risk resilience.

According to Nguyen.T.T (2023), who examined the factors affecting the financial self-sustainability (FSS) of microfinance institutions, operating income has a positive relationship with financial sustainability. Therefore, author expects that operating income has a positive relationship with the financial sustainability of banks. The formula for INC is as follows:

$$\ln\text{INC} = \text{Natural log of income operation}$$

**Hypothesis H6:** Operating income ( $\ln\text{INC}$ ) has a *positive impact* on the financial sustainability of Vietnamese commercial banks.

**Loan-to-Deposit Ratio (LTD):** Loan-to-Deposit Ratio (LTD): Hesse and Čihák (2007) highlighted an inverse relationship between the loan-to-deposit ratio and a bank's financial stability. Specifically, banks with a high LDR, meaning that a significant portion of their assets is allocated to lending activities, tend to face an increased risk of non-performing loans. This leads to higher credit risk, weakening profitability and negatively impacting long-term financial sustainability. An ideal LTD ratio is recommended to be in the range of 80-100%. To ensure long-term sustainability in the current volatile environment and to encourage safe lending practices, author expects that LTD has a negative relationship with financial sustainability. The formula for LTD is as follows:

$$\text{LTD} = \frac{\text{Total Loans}}{\text{Total Deposits}}$$

**Hypothesis H7:** The Loan-to-Deposit Ratio (LTD) has a *negative impact* on the financial sustainability of Vietnamese commercial banks.

**Board Independence Ratio (BI) :** The presence of independent board members plays a critical role in ensuring bank stability. Due to their objectivity and lack of direct ties to executive management or major shareholders, independent directors act as key overseers, providing unbiased decision-making, enhancing operational efficiency, and minimizing risks.

Numerous global financial scandals have demonstrated the consequences of weak governance and the absence of independent board supervision. Although no specific studies have directly examined the relationship between board independence and the financial sustainability of Vietnamese commercial banks, this study anticipates a positive correlation between these two factors. The formula for BI is as follows:

$$\mathbf{BI} = \frac{\text{The number of independent members}}{\text{The number of members on the board}}$$

**Hypothesis H8:** The Board Independence Ratio (BI) has a *positive impact* on the financial sustainability of Vietnamese commercial banks.

**Economic Growth Rate (GDP):** Research by Gupta & Kashiramka (2020) and Rupeika-Apoga et al. (2018) suggests that GDP growth has a positive relationship with bank financial stability. Specifically, higher GDP growth stimulates business activities and consumer spending, leading to increased credit demand, which strengthens financial stability in banks.

**Hypothesis H9:** GDP Growth Rate (GDP) has a *positive impact* on the financial sustainability of Vietnamese commercial banks.

**Inflation rate (INF):** Inflation exerts a significant influence on the economy, particularly in the banking sector. High inflation leads to currency depreciation, reducing the real value of both loans and assets held by banks. This increases credit risk as borrowers' repayment capacity declines. Empirical studies by Lan (2021) and Rahim et al. (2012) indicate that inflation has a negative impact on financial sustainability.

**Hypothesis H10:** Inflation rate (INF) has a *negative impact* on the financial sustainability of Vietnamese commercial banks.

**Table 3. 1: Summary of variables and measurement methods in the model**

| Type of Variables            | Variables                                    | Symbol | Measurement Formula                                                              | Expected sign | Prior studies                                                         |
|------------------------------|----------------------------------------------|--------|----------------------------------------------------------------------------------|---------------|-----------------------------------------------------------------------|
| <b>Dependent variables</b>   | Financial sustainability of Commercial Banks | FSR    | $\text{FSR} = \frac{\text{Total finance income}}{\text{Total finance expenses}}$ |               | <i>Sanfa &amp; Ida (2023); Arifani (2020); Nguyen Thi Thuy (2023)</i> |
| <b>Independent variables</b> | Bank size                                    | lnSIZE | lnSIZE = Natural log of total assets                                             | -             | <i>Daoud &amp; Kammoun (2020)</i>                                     |
|                              | Return on assets                             | ROA    | $\text{ROA} = \frac{\text{Net Profit After Tax}}{\text{Total assets}}$           | +             | <i>Meher &amp; Getaneh (2019); Moechdi et</i>                         |

|  |                              |     |                                                                                  |   |                                                                                                                          |
|--|------------------------------|-----|----------------------------------------------------------------------------------|---|--------------------------------------------------------------------------------------------------------------------------|
|  |                              |     |                                                                                  |   | <i>al. (2016);<br/>Shakhrun<br/>et al.,<br/>(2019);<br/>Sanfa &amp;<br/>Ida (2023)</i>                                   |
|  | Capital<br>Adequacy<br>Ratio | CAR | $\text{CAR} = \frac{\text{Equity capital}}{\text{Risk – weighted assets}}$       | + | <i>Fajari &amp;<br/>Sunarto<br/>(2017); Ayu<br/>ningtyas et<br/>al. (2018);<br/>Munandar<br/>&amp; Aravik<br/>(2022)</i> |
|  | Non<br>performing<br>Loan    | NPL | $\text{NPL} = \frac{\text{Group Loans (3,4,5)}}{\text{Total outstanding loans}}$ | - | <i>Rajan &amp;<br/>Dhal<br/>(2003);<br/>Mabruroh<br/>(2004)</i>                                                          |
|  | Liquidity<br>Ratio           | LIQ | $\text{LIQ} = \frac{\text{Current assets}}{\text{Total assets}}$                 | + | <i>Rajhi &amp;<br/>Hassairi<br/>(2013)</i>                                                                               |

|  |                          |       |                                                                                                   |   |                                                                   |
|--|--------------------------|-------|---------------------------------------------------------------------------------------------------|---|-------------------------------------------------------------------|
|  | Operating Income         | lnINC | lnINC = Natural log of income operation                                                           | + | <i>Nguyen Thi Thuy (2023)</i>                                     |
|  | Loan to Deposit Ratio    | LTD   | $LTD = \frac{\text{Total Loans}}{\text{Total Deposits}}$                                          | - | <i>Hesse and Čihák (2007)</i>                                     |
|  | Board Independence Ratio | BI    | BI = $\frac{\text{The number of independent members}}{\text{The number of members on the board}}$ | + | <i>N/A</i>                                                        |
|  | Economic Growth          | GDP   | Growth rate of t year                                                                             | + | <i>Fupta &amp; Kashiramka (2020); Rupeika-Apoga et al. (2018)</i> |
|  | Inflation rate           | INF   | Inflation of t year                                                                               | - | <i>Lan (2021); Rahim et al. (2012)</i>                            |

*Source: Compiled by the author*

### 3.5. RESEARCH PROCESS

More specifically, the research process is presented by the author in Table 3.2

**Table 3. 2: Research process**

| Step | Description                                                       |
|------|-------------------------------------------------------------------|
| 1    | Identification of the Research Topic and Rationale                |
| 2    | Researching and Analyzing the Related Scientific Studies          |
| 3    | Development of a Research Foundation and Work Plan                |
| 4    | Determination of Research Scope, Methodology, and Data Collection |
| 5    | Data Examination, Refinement, and Organization                    |
| 6    | Data Processing and Statistical Analysis using Stata 17.0         |
| 7    | Model Estimation using Pooled OLS, FEM, and REM                   |
| 8    | Selection of the Most Suitable Model                              |
| 9    | Diagnostic Tests and Correction for Model Deficiencies            |
| 10   | Conclusion and Policy Recommendations                             |

*Source: Compiled by the author.*

## **SUMMARY OF CHAPTER 3**

Based on the general theoretical framework and prior studies, develops a research model to examine the impact of independent variables on the financial sustainability of banks. Specifically, the author proposes research hypotheses for each independent variable, thereby determining the expected direction of their effects within the model. To ensure rigor and systematic analysis, the author presents the measurement methods for each variable, including quantitative criteria and specific calculation methods, ensuring objectivity and data reliability. The data processing procedure is also outlined, covering steps such as data cleaning, aggregation, and standardization to maintain consistency before being incorporated into the analytical model.

Regarding research methodology, the author employs the System Generalized Method of Moments (SGMM) regression model to address potential endogeneity issues within the model. Additionally, Quantile Regression is utilized to assess the impact of independent variables across different quantiles of the dependent variable. This approach allows for a deeper analysis of how various factors influence the financial sustainability of banks at different levels. The combination of these two methods enhances the reliability and comprehensiveness of the research findings. SGMM helps control endogeneity issues and dynamic relationships between variables, while Quantile Regression provides insights into the varying degrees of influence exerted by independent variables across different bank groups. This comprehensive analysis offers valuable insights into the determinants of financial sustainability in banks, thereby providing policy recommendations to improve the efficiency and resilience of the banking system in an increasingly volatile economic environment.

## CHAPTER 4. RESEARCH RESULTS

### 4.1. DESCRIPTIVE STATISTICAL ANALYSIS

**Table 4. 1: Descriptive statistical results**

| <b>Variable</b> | <b>Obs</b> | <b>Mean</b> | <b>Std. dev.</b> | <b>Min</b> | <b>Max</b> |
|-----------------|------------|-------------|------------------|------------|------------|
| <b>FSR</b>      | 378        | 1.1983      | 0.1821           | 0.8203     | 2.0156     |
| <b>lnSIZE</b>   | 378        | 18.7575     | 1.24700          | 15.9227    | 21.5565    |
| <b>ROA</b>      | 378        | 0.0092      | 0.0088           | -0.0551    | 0.0998     |
| <b>CAR</b>      | 378        | 0.1337      | 0.04142          | 0.0835     | 0.3342     |
| <b>NPL</b>      | 378        | 0.0232      | 0.02833          | 0.00018    | 0.4029     |
| <b>LIQ</b>      | 378        | 0.92605     | 0.04637          | 0.7538     | 0.9918     |
| <b>LnINC</b>    | 378        | 14.3107     | 1.76632          | 10.2659    | 18.5930    |
| <b>LTD</b>      | 378        | 0.8711      | 0.18144          | 0.0236     | 1.5599     |
| <b>BI</b>       | 378        | 0.1395      | 0.08243          | 0,000      | 0.4        |
| <b>GDP</b>      | 378        | 0.0601      | 0.01546          | 0.026      | 0.08       |
| <b>INF</b>      | 378        | 0.0516      | 0.04465          | 0.006      | 0.1868     |

*Source: Compiled by the author using Stata 17 software.*

The results in the table above illustrate the financial sustainability of Vietnamese commercial banks during the 2010–2023 period, measured by the Financial Sustainability Ratio (FSR), which has an average value of 119.83% and a standard deviation of 18.21%. Among them, Loc Phat Commercial Joint Stock Bank (Ticker: LPB) recorded the highest financial sustainability in 2010, reaching

201.56%. According to Arifani (2020), Loc Phat Bank was considered financially sustainable as its FSR exceeded 100%. It can be observed that Loc Phat Bank had a relatively high sustainability ratio in the early years compared to other banks due to its smaller initial scale, which made it less affected by external factors compared to larger banks. Conversely, the lowest FSR of 82.03% was recorded by Tien Phong Commercial Joint Stock Bank (Ticker: TPB) in 2011.

The average bank size (SIZE) was 18.7575, with a standard deviation of 1.2470 during the 2010–2023 period. The smallest bank size was 15.9227, recorded by BV Commercial Joint Stock Bank (Ticker: BVB) in 2010, while the largest bank size was 21.5565, recorded by Vietnam Investment and Development Commercial Joint Stock Bank (Ticker: BID) in 2023.

The average return on assets (ROA) was 0.0092, with a standard deviation of 0.0088 over the same period. The lowest ROA was -0.0551, recorded by Tien Phong Bank (Ticker: TPB) in 2011, while the highest ROA was 0.0998, recorded by the Vietnam Bank for Agriculture and Rural Development (Ticker: AGR) in 2023.

The average capital adequacy ratio (CAR) was 0.1337, with a standard deviation of 0.0414. The lowest CAR was 0.0835, recorded by Loc Phat Bank (Ticker: LPB) in 2019, whereas the highest CAR was 0.3342, recorded by Kien Long Commercial Joint Stock Bank (Ticker: KLB) in 2012.

The average non-performing loan (NPL) ratio was 0.02328, with a standard deviation of 0.02833. The lowest NPL ratio was 0.00018, recorded by Tien Phong Bank (Ticker: TPB) in 2010, while the highest NPL ratio was 0.4029, recorded by BV Bank (Ticker: BVB) in 2022.

The average liquidity ratio (LIQ) was 0.92605, with a standard deviation of 0.04637. The lowest LIQ was 0.7538, recorded by Nam A Commercial Joint Stock Bank (Ticker: NAB) in 2011, while the highest LIQ was 0.9918, recorded by Vietnam Export-Import Commercial Joint Stock Bank (Ticker: EIB) in 2010.

The average operating income (INC) was 14.3107, with a standard deviation of 1.76632. The lowest INC was 10.2659, recorded by Viet A Commercial Joint Stock Bank (Ticker: VAB) in 2012, while the highest INC was 18.5930, recorded by Investment and Development Bank (Ticker: BID) in 2023.

The average loan-to-deposit ratio (LTD) was 0.8711, with a standard deviation of 0.18144. The lowest LTD was 0.0236, recorded by Maritime Commercial Joint Stock Bank (Ticker: MSB) in 2016, while the highest LTD was 1.5599, recorded by Vietnam Prosperity Commercial Joint Stock Bank (Ticker: VPB) in 2023.

The average independent board members ratio (BI) was 0.13959, with a standard deviation of 0.0824. The lowest BI value was consistently recorded by the Vietnam Bank for Agriculture and Rural Development (Ticker: AGR), which appeared to be close to zero, whereas the highest BI value was 0.4000, recorded by BV Bank (Ticker: BVB) in 2022.

During the 2010–2023 period, the average economic growth rate (GDP) was 6.015%, with a standard deviation of 1.546%. The lowest GDP growth rate was 2.6% (2021), while the highest was 8.001% (2022). The average inflation rate (INF) was 5.16%, with a standard deviation of 4.46%. The lowest and highest inflation rates were 0.6% (2015) and 18.68% (2011), respectively.

## **4.2. CORRELATION MATRIX**

To evaluate the strength and direction of the linear relationship between the variables in this study, the author conducts a correlation analysis. The correlation coefficient, which ranges from -1 to +1, quantifies this relationship: values approaching  $\pm 1$  indicate a strong correlation, negative values signify an inverse relationship, positive values reflect a direct relationship, and a coefficient of 0 suggests no linear association. The results of this analysis, including the statistical significance of the correlation coefficients, are detailed in the table below.

### Table 4. 2: Correlation matrix

|                                                                                                                                  | LTD        | LnINC      | BI        | GDP        | INF       |            |
|----------------------------------------------------------------------------------------------------------------------------------|------------|------------|-----------|------------|-----------|------------|
|                                                                                                                                  |            |            |           |            |           |            |
|                                                                                                                                  | FSR        | lnSIZE     | ROA       | CAR        | NPL       | LIQ        |
| FSR                                                                                                                              | 1          |            |           |            |           |            |
| lnSIZE                                                                                                                           | 0.1494***  | 1          |           |            |           |            |
| ROA                                                                                                                              | 0.5433***  | 0.2575***  | 1         |            |           |            |
| CAR                                                                                                                              | 0.0622     | -0.5850*** | 0.0161    | 1          |           |            |
| NPL                                                                                                                              | -0.1075 ** | -0.1028**  | -0.1318** | 0.0317     | 1         |            |
| LIQ                                                                                                                              | 0.3046***  | 0.4465***  | 0.2298*** | -0.1585*** | -0.1271** | 1          |
| lnINC                                                                                                                            | 0.4780***  | 0.8315***  | 0.3604*** | -0.4102*** | -0.0641   | 0.4416***  |
| LTD                                                                                                                              | 0.1957***  | 0.2758***  | 0.2875*** | -0.0403    | -0.0353   | 0.3862***  |
| BI                                                                                                                               | -0.1702*** | -0.0987*   | -0.0444   | -0.0114    | 0.0302    | -0.0475    |
| GDP                                                                                                                              | -0.0225    | -0.0947*   | -0.0882*  | 0.0915*    | 0.0486    | -0.0337    |
| INF                                                                                                                              | 0.0896*    | -0.2894*** | 0.0240    | 0.2840***  | 0.0650    | -0.2802*** |
| <i>Note: *, **, ***: The coefficients are statistically significant at the 10%, 5% and 1% significance levels, respectively.</i> |            |            |           |            |           |            |

|                                                                                                                                  |            |            |            |        |   |
|----------------------------------------------------------------------------------------------------------------------------------|------------|------------|------------|--------|---|
| <b>FSR</b>                                                                                                                       |            |            |            |        |   |
| <b>lnSIZE</b>                                                                                                                    |            |            |            |        |   |
| <b>ROA</b>                                                                                                                       |            |            |            |        |   |
| <b>CAR</b>                                                                                                                       |            |            |            |        |   |
| <b>NPL</b>                                                                                                                       |            |            |            |        |   |
| <b>LIQ</b>                                                                                                                       |            |            |            |        |   |
| <b>lnINC</b>                                                                                                                     | 1          |            |            |        |   |
| <b>LTD</b>                                                                                                                       | 0.3028***  | 1          |            |        |   |
| <b>BI</b>                                                                                                                        | -0.1424*** | 0.1171**   | 1          |        |   |
| <b>GDP</b>                                                                                                                       | -0.0952*   | -0.0969*   | -0.0460    | 1      |   |
| <b>INF</b>                                                                                                                       | -0.1673*** | -0.1537*** | -0.2712*** | 0.0394 | 1 |
| <i>Note: *, **, ***: The coefficients are statistically significant at the 10%, 5% and 1% significance levels, respectively.</i> |            |            |            |        |   |

*Source: Compiled by the author using Stata 17 software.*

The correlation matrix above allows us to assess the linear relationship between the dependent variable and independent variables, as well as the relationships among independent variables to identify potential multicollinearity.

In economic research, an absolute correlation threshold of 0.8 is commonly used as a benchmark. When the correlation coefficient between two variables exceeds this threshold, it indicates a strong linear relationship, which may lead to multicollinearity issues. The results in Table 4.2 show that all correlation values remain within acceptable limits, as none exceed  $\pm 0.8$ .

Among the variables, NPL, BI, and GDP exhibit a negative correlation with the dependent variable, represented by the FSR index, while lnSIZE, ROA, CAR,

LIQ, lnINC, LTD, and INF demonstrate a positive correlation with it. These relationships suggest that financial sustainability as measured by the FSR index, is influenced in different directions depending on the nature of the independent variables.

#### 4.3. MULTICOLLINEARITY TEST (VIF TEST)

To assess the presence of multicollinearity in the research model, the Variance Inflation Factor (VIF) test is conducted using Stata. A VIF value below 10 indicates that multicollinearity is not a concern, ensuring the independence of explanatory variables. However, if the VIF value exceeds 10, it suggests a high degree of multicollinearity, which may distort coefficient estimates and affect the reliability of the regression model.

**Table 4. 3: VIF test results**

| <b>Variable</b> | <b>VIF</b> | <b>1/VIF</b> |
|-----------------|------------|--------------|
| <b>lnSIZE</b>   | 4.53       | 0.220662     |
| <b>lnINC</b>    | 3.69       | 0.270646     |
| <b>CAR</b>      | 1.69       | 0.673075     |
| <b>LIQ</b>      | 1.49       | 0.673075     |
| <b>INF</b>      | 1.32       | 0.759000     |
| <b>LTD</b>      | 1.31       | 0.765193     |
| <b>ROA</b>      | 1.27       | 0.788361     |
| <b>BI</b>       | 1.17       | 0.856657     |
| <b>NPL</b>      | 1.04       | 0.960373     |
| <b>GDP</b>      | 1.03       | 0.973980     |
| <b>Mean VIF</b> | 1.85       |              |

*Source: Compiled by the author using Stata 17 software.*

Based on the results in Table 4.3, which displays the Variance Inflation Factor (VIF) test, the highest VIF value recorded is 4.53 (lnSIZE), while the lowest is 1.03

(GDP). Since all values remain well below the critical threshold of 10, it can be inferred that multicollinearity is not an issue in this model, and no independent variables need to be removed.

This outcome confirms that the predictor variables do not exhibit excessive interdependence, preserving the reliability of the regression estimates. Therefore, the model satisfies the multicollinearity assumption, ensuring that all variables can be retained for further analysis. This allows for a more robust and accurate evaluation of the relationships between the independent and dependent variables.

#### **4.4. REGRESSION RESULTS OF PANEL DATA MODELS: POOLED OLS, FEM, AND REM**

The author conducts panel data regression analysis based on the collected data using Stata 17.0. Three estimation methods are applied: Pooled Ordinary Least Squares (Pooled OLS), Fixed Effects Model (FEM), and Random Effects Model (REM) to assess the impact of 10 independent variables on the dependent variable. The regression results are summarized in the following table.

**Table 4. 4: Regression Results of Pooled OLS, FEM, and REM Models**

| <b>Dependent variable</b>   | <b>FSR</b>         |                |                    |                |                    |                |
|-----------------------------|--------------------|----------------|--------------------|----------------|--------------------|----------------|
|                             | <b>Pooled OLS</b>  |                | <b>FEM</b>         |                | <b>REM</b>         |                |
| <b>Independent variable</b> | <b>Coefficient</b> | <b>P-value</b> | <b>Coefficient</b> | <b>P-value</b> | <b>Coefficient</b> | <b>P-value</b> |
| <b>lnSIZE</b>               | -0.1158117***      | 0.000          | -0.1243101***      | 0.000          | -0.1243656***      | 0.000          |
| <b>ROA</b>                  | 7.674991***        | 0.000          | 7.40222***         | 0.000          | 7.55227***         | 0.000          |
| <b>CAR</b>                  | -0.0653835         | 0.731          | 0.0297344          | 0.882          | -0.0156791         | 0.934          |
| <b>NPL</b>                  | -0.3889531*        | 0.075          | -0.3309922         | 0.107          | -0.3459586*        | 0.085          |

|                                                                                                                                         |                                              |       |                                                                       |       |                                                                       |       |
|-----------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|-------|-----------------------------------------------------------------------|-------|-----------------------------------------------------------------------|-------|
| <b>LIQ</b>                                                                                                                              | 0.6776061***                                 | 0.000 | 0.5336593***                                                          | 0.005 | 0.5470449***                                                          | 0.002 |
| <b>lnINC</b>                                                                                                                            | 0.0950282***                                 | 0.000 | 0.1099211***                                                          | 0.000 | 0.1051727***                                                          | 0.000 |
| <b>LTD</b>                                                                                                                              | -0.0242611                                   | 0.525 | 0.033907                                                              | 0.415 | 0.0224171                                                             | 0.560 |
| <b>BI</b>                                                                                                                               | -0.170094**                                  | 0.033 | -0.2468979***                                                         | 0.007 | -0.2162099***                                                         | 0.009 |
| <b>GDP</b>                                                                                                                              | 0.3045667                                    | 0.443 | 0.3780952                                                             | 0.282 | 0.3563506                                                             | 0.307 |
| <b>INF</b>                                                                                                                              | 0.147758                                     | 0.343 | 0.0900739                                                             | 0.568 | 0 .0916323                                                            | 0.531 |
| <b>_cons</b>                                                                                                                            | 1.349054***                                  | 0.000 | 1.37569***                                                            | 0.000 | 1.444211***                                                           | 0.000 |
| <b>N</b>                                                                                                                                | 378                                          | 378   | 378                                                                   | 378   | 378                                                                   | 378   |
|                                                                                                                                         | R-squared = 0.5940<br>Adj R-squared = 0.5829 |       | R-squared:<br>Within = 0.5784<br>Between = 0.5950<br>Overall = 0.5873 |       | R-squared:<br>Within = 0.5777<br>Between = 0.6056<br>Overall = 0.5907 |       |
| <i><b>Note:</b> *, **, ***: The coefficients are statistically significant at the 10%, 5% and 1% significance levels, respectively.</i> |                                              |       |                                                                       |       |                                                                       |       |

*Source: Compiled by the author using Stata 17 software.*

**Pooled OLS Model:** The R-squared value of 0.5940 indicates that the independent variables explain 59.40% of the variation in financial sustainability. ROA, LIQ, and lnINC have a positive impact on financial sustainability, while lnSIZE, NPL, and BI exhibit a negative influence. Additionally, CAR, LTD, GDP, and INF are statistically insignificant, suggesting minimal impact when time factors or firm-specific characteristics are excluded.

**FEM Model:** The R-squared value of 0.5784 suggests that independent variables account for 57.84% of the variation in financial sustainability. ROA, LIQ,

and lnINC positively influence financial sustainability, whereas BI and lnSIZE have a negative effect. Furthermore, CAR, NPL, LTD, GDP, and INF are statistically insignificant, implying minimal influence when time effects or firm-specific attributes are omitted.

**REM Model:** With an R-squared value of 0.5777, the independent variables explain 57.77% of the variation in financial sustainability. ROA, LIQ, and lnINC positively contribute to financial sustainability, while lnSIZE, NPL, and BI have a negative impact. Moreover, CAR, LTD, GDP, and INF are statistically insignificant, indicating a negligible effect when time-related or firm-specific characteristics are disregarded.

## 4.5. SELECTING THE MOST SUITABLE REGRESSION MODEL AMONG POOLED OLS, FEM, AND REM

### 4.5.1 *Selection between Pooled OLS and FEM*

For the dependent variable represented by the FRS index, the author employs the F-test to determine the appropriate model between Pooled OLS and FEM. The hypotheses are formulated as follows:

$H_0$ : The Pooled OLS model is appropriate.

$H_1$ : The FEM model is appropriate.

**Table 4. 5: Regression Results Between Pooled OLS and FEM**

| F-test | P-value                 |
|--------|-------------------------|
|        | Prob > chibar2 = 0.0000 |

*Source: Compiled by the author using Stata 17 software.*

The test results indicate that the P-value is  $0.0000 < 5\%$ , leading to the rejection of the null hypothesis  $H_0$ . Therefore, the FEM model is more appropriate than the Pooled OLS model.

### 4.5.2 *Selection between Pooled OLS and REM*

For the dependent variable represented by the FRS index, the author conducts the Breusch and Pagan Lagrangian test to determine the appropriate model between Pooled OLS and REM. The hypotheses are formulated as follows:

$H_0$ : The Pooled OLS model is appropriate.

$H_1$ : The REM model is appropriate.

**Table 4. 6: Regression Results Between Pooled OLS and REM**

|                              |                         |
|------------------------------|-------------------------|
| Breusch and Pagan Lagrangian | P-value                 |
|                              | Prob > chibar2 = 0.0000 |

*Source: Compiled by the author using Stata 17 software.*

The test results indicate that the P-value is  $0.0000 < 5\%$ , leading to the rejection of the null hypothesis  $H_0$ . Therefore, the REM model is more appropriate than the Pooled OLS model.

#### **4.5.3. Selection between FEM and REM**

For the dependent variable represented by the FRS index, the author conducts the Hausman test to determine the appropriate model between FEM and REM. The hypotheses are formulated as follows:

$H_0$ : The REM model is appropriate.

$H_1$ : The FEM model is appropriate.

**Table 4. 7: Regression Results between REM and FEM**

|         |                         |
|---------|-------------------------|
| Hausman | P-value                 |
|         | Prob > chibar2 = 0.5678 |

*Source: Compiled by the author using Stata 17 software.*

The test results indicate that the P-value is  $0.5678 > 5\%$ , leading to the acceptance of the null hypothesis  $H_0$ . Therefore, the REM model is more appropriate than the FEM model.

## Conclusion

After conducting the F-test, Breusch-Pagan test, and Hausman test to determine the most suitable regression model, the author concludes that the REM model is the most appropriate when the dependent variable is represented by the FSR index. The author will proceed with diagnostic tests based on the regression results of the REM model.

### 4.6. TESTING FOR MODEL VIOLATIONS

#### 4.6.1. *Heteroskedasticity test*

To examine the presence of heteroskedasticity in the REM model, the author employs the Breusch-Pagan LM test. The hypotheses for this test are formulated as follows:

$H_0$ : The model does not exhibit heteroskedasticity.

$H_1$ : The model exhibits heteroskedasticity.

**Table 4. 8: Breusch-Pagan LM Test Results**

| Breusch and Pagan Lagrangian | P-value                 |
|------------------------------|-------------------------|
|                              | Prob > chibar2 = 0.0000 |

*Source: Compiled by the author using Stata 17 software.*

After applying the Breusch-Pagan LM test in Table 4.8, the results show that  $\text{Prob} > \text{chibar2} = 0.0000 < 5\%$ . This leads to the rejection of the null hypothesis ( $H_0$ ) and acceptance of the alternative hypothesis ( $H_1$ ), indicating the presence of heteroscedasticity in the model.

#### 4.6.2. Autocorrelation test

Following the heteroscedasticity test, the author proceeds to examine the presence of autocorrelation in the model using the Wooldridge test. The hypotheses are formulated as follows:

$H_0$ : The model does not exhibit autocorrelation.

$H_1$ : The model exhibits autocorrelation.

**Table 4. 9: Wooldridge Test Results**

| Wooldridge | P-value                 |
|------------|-------------------------|
|            | Prob > chibar2 = 0.0000 |

*Source: Compiled by the author using Stata 17 software.*

After applying the Wooldridge test in Table 4.9, the results indicate that  $\text{Prob} > \text{chibar2} = 0.0000 < 5\%$ . This leads to the rejection of the null hypothesis ( $H_0$ ) and acceptance of the alternative hypothesis ( $H_1$ ), confirming the presence of autocorrelation in the model.

#### 4.7. ADDRESSING THE VIOLATIONS OF THE SELECTED MODEL

After testing for model violations, the results reveal that the REM regression model suffers from both heteroscedasticity and autocorrelation issues. To address these problems, the author employs the Feasible Generalized Least Squares (FGLS) method. The results of the FGLS estimation are presented in the table below.

**Table 4. 10: FGLS Model Results**

| Independent variable                                                                                                             | Dependent variable FSR |         |
|----------------------------------------------------------------------------------------------------------------------------------|------------------------|---------|
|                                                                                                                                  | Coefficient            | P-value |
| lnSIZE                                                                                                                           | -0.1272723***          | 0.000   |
| ROA                                                                                                                              | 7.781971***            | 0.000   |
| CAR                                                                                                                              | -0.1222964             | 0.511   |
| NPL                                                                                                                              | -0.3093238*            | 0.069   |
| LIQ                                                                                                                              | 0.4958366***           | 0.001   |
| lnINC                                                                                                                            | 0.0941865***           | 0.000   |
| LTD                                                                                                                              | -0.0259047             | 0.469   |
| BI                                                                                                                               | -0.242235***           | 0.000   |
| GDP                                                                                                                              | 0.5773549**            | 0.018   |
| INF                                                                                                                              | -0.4424776***          | 0.000   |
| C                                                                                                                                | 1.775196***            | 0.000   |
| Number of obs                                                                                                                    | 376                    |         |
| Number of groups                                                                                                                 | 27                     |         |
| Wald chi2(10)                                                                                                                    | 609.89                 |         |
| Prob > chi2                                                                                                                      | 0.0000                 |         |
| <i>Note: *, **, ***: The coefficients are statistically significant at the 10%, 5% and 1% significance levels, respectively.</i> |                        |         |

*Source: Compiled by the author using Stata 17 software.*

The model is constructed using the Feasible Generalized Least Squares (FGLS) method with 376 observations over a 14-year period. It is considered

statistically significant, as indicated by the Prob > chi2 value of 0.0000, which is below the 5% threshold. Therefore, this model is deemed appropriate for the study.

From Table 4.10, it can be observed that most variables in the model are statistically significant, except for CAR and LTD, as their p-values exceed the 1%, 5%, and 10% significance levels. Among the variables, lnSIZE, NPL, BI, and INF exhibit a negative relationship with the dependent variable FSR, while ROA, LIQ, lnINC, and GDP have a positive association with FSR.

#### 4.8. TESTING FOR ENDOGENEITY AND THE SGMM METHOD

##### 4.8.1. *Endogeneity testing*

Although the FGLS method has been applied to correct for autocorrelation and heteroskedasticity, there remains the possibility of endogenous variables that this approach cannot fully address. Therefore, the author employs the Durbin-Wu-Hausman test to assess the endogeneity of each independent variable in the model. The test results are presented in Table 4.11.

**Table 4. 11: Summary of Endogenous and Exogenous Variables in the Model**

| Variable | Durbin-Wu-Hausman test with hypotheses:<br><br>- H <sub>0</sub> : The model contains variable X as an exogenous variable.<br>- H <sub>1</sub> : The model contains variable X as an endogenous variable. | Conclusion                                                   |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| lnSIZE   | $p = 0.0223 < 0.05$                                                                                                                                                                                      | Accepted H <sub>1</sub> : The variable lnSIZE is endogenous. |

|              |                     |                                                                           |
|--------------|---------------------|---------------------------------------------------------------------------|
| <b>ROA</b>   | $p = 0.0001 < 0.05$ | <b>Accepted <math>H_1</math>:</b> The variable <b>ROA</b> is endogenous.  |
| <b>CAR</b>   | $p = 0.9082 > 0.05$ | <b>Accepted <math>H_0</math>:</b> The variable <b>CAR</b> is exogenous.   |
| <b>NPL</b>   | $p = 0.6580 > 0.05$ | <b>Accepted <math>H_0</math>:</b> The variable <b>NPL</b> is exogenous.   |
| <b>LIQ</b>   | $p = 0.0022 < 0.05$ | <b>Accepted <math>H_1</math>:</b> The variable <b>LIQ</b> is endogenous.  |
| <b>lnINC</b> | $p = 0.8062 > 0.05$ | <b>Accepted <math>H_0</math>:</b> The variable <b>lnINC</b> is exogenous. |
| <b>LTD</b>   | $p = 0.7496 > 0.05$ | <b>Accepted <math>H_0</math>:</b> The variable <b>LTD</b> is exogenous.   |
| <b>BI</b>    | $p = 0.1466 > 0.05$ | <b>Accepted <math>H_0</math>:</b> The variable <b>BI</b> is exogenous.    |
| <b>GDP</b>   | $p = 0.0001 < 0.05$ | <b>Accepted <math>H_1</math>:</b> The variable <b>GDP</b> is endogenous.  |
| <b>INF</b>   | $p = 0.4728 > 0.05$ | <b>Accepted <math>H_0</math>:</b> The variable <b>INF</b> is exogenous.   |

*Source: Compiled by the author using Stata 17 software.*

#### 4.8.2. SGMM model

To address the endogeneity issue identified in section 4.8.1, the System Generalized Method of Moments (SGMM) is employed. SGMM is an advanced econometric technique that effectively handles endogeneity by utilizing internal instruments derived from lagged values of the endogenous variables.

The estimation results from the SGMM model are presented in Table 4.12, which provides insights into the impact of independent variables on financial sustainability while mitigating bias from endogenous regressors.

**Table 4. 12: SGMM Model Results**

| Variable                      | FSR           |         |
|-------------------------------|---------------|---------|
|                               | Coefficient   | P-value |
| <b>FSR L1.</b>                | 0.407201***   | 0.000   |
| <b>lnSIZE</b>                 | -0.0715745*** | 0.000   |
| <b>ROA</b>                    | 5.359878***   | 0.002   |
| <b>CAR</b>                    | -0.1048387    | 0.653   |
| <b>NPL</b>                    | -0.9797766*   | 0.078   |
| <b>LIQ</b>                    | -0.9407358    | 0.140   |
| <b>lnINC</b>                  | 0.0650332***  | 0.001   |
| <b>LTD</b>                    | 0.1775695**   | 0.035   |
| <b>BI</b>                     | 0.2647368*    | 0.094   |
| <b>GDP</b>                    | 0.4732932**   | 0.022   |
| <b>INF</b>                    | -2.159927***  | 0.001   |
| <b>C</b>                      | 1.822373**    | 0.010   |
| <b>Prob &gt; F = 0.000</b>    |               |         |
| <b>Number of groups</b>       | 27            |         |
| <b>Number of observations</b> | 214           |         |

|                                                                                                                                         |                     |
|-----------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| <b>Number of instruments</b>                                                                                                            | 27                  |
| <b>AR1 test</b>                                                                                                                         | Pr > z = 0.070      |
| <b>AR2 test</b>                                                                                                                         | Pr > z = 0.649      |
| <b>Sargan test</b>                                                                                                                      | Prob > chi2 = 0.106 |
| <b>Hansen test</b>                                                                                                                      | Prob > chi2 = 0.102 |
| <i><b>Note:</b> *, **, ***: The coefficients are statistically significant at the 10%, 5% and 1% significance levels, respectively.</i> |                     |

*Source: Compiled by the author using Stata 17 software.*

The regression results using the SGMM method are presented in Table 4.12. The author observes that, in all models, the number of instrumental variables does not exceed the number of groups. In the Arellano-Bond test, all Pr > z values for AR(1) and AR(2) are greater than 5%. Regarding the Sargan test, the Prob > chi2 values exceed 10%, and the same applies to the Hansen test. The SGMM regression model satisfies all four aforementioned conditions, indicating that the model presented by the author is appropriate, efficient, and highly accurate. Therefore, the author employs the SGMM model for result discussion.

#### **4.9. SUMMARY AND DISCUSSION OF RESEARCH RESULTS**

Despite the inherent differences among Vietnamese commercial banks, this study temporarily abstracts from those variations to focus on evaluating the impact of key factors in the research model on financial sustainability (FSR). Table 4.13 presents the findings on how these factors influence the financial sustainability of Vietnamese commercial banks, utilizing the SGMM method.

To capture the effects of explanatory variables across different levels of financial sustainability, the study employs quantile regression at five different quantiles: the 10th percentile (Q10), 25th percentile (Q25), median or 50th percentile (Q50), 75th percentile (Q75), and 90th percentile (Q90). This approach allows for a more comprehensive understanding of how the determinants of financial

sustainability may vary across banks with lower, average, and higher levels of FSR, rather than just analyzing the mean effects as in traditional regression models.

**Table 4. 13: Results of the SGMM Method with Quantile Regression at Different Percentiles**

| Variable      | SGMM         | Q10      | Q25      | Q50        | Q75       | Q90       |
|---------------|--------------|----------|----------|------------|-----------|-----------|
| <b>FSR</b>    | 0.407201***  |          |          |            |           |           |
|               | [4.12]       |          |          |            |           |           |
| <b>lnSIZE</b> | -0.071574*** | -0.0300  | -0.0317  | -0.0890*** | -0.106*** | -0.128*** |
|               | [-4.75]      | [-1.09]  | [-1.10]  | [-4.00]    | [-9.57]   | [-9.92]   |
| <b>ROA</b>    | 5.359878***  | 8.419**  | 12.69*** | 11.16***   | 9.810***  | 10.88***  |
|               | [3.38]       | [2.55]   | [3.68]   | [4.58]     | [4.13]    | [5.18]    |
| <b>CAR</b>    | -0.104838    | -0.181   | -0.277   | -0.143     | 0.391**   | 0.199     |
|               | [-0.45]      | [-0.89]  | [-1.35]  | [-0.56]    | [1.97]    | [0.49]    |
| <b>NPL</b>    | -0.979776*   | -0.142   | -0.337   | -0.0879    | -0.210    | -0.406    |
|               | [-1.83]      | [-0.22]  | [-0.86]  | [-0.18]    | [-0.83]   | [-0.85]   |
| <b>LIQ</b>    | -0.9407358   | 0.476**  | 0.385**  | 0.705***   | 0.662***  | 0.535*    |
|               | [-1.52]      | [2.16]   | [2.41]   | [3.25]     | [3.25]    | [1.81]    |
| <b>lnINC</b>  | 0.0650332*** | 0.0267   | 0.0227   | 0.0614***  | 0.101***  | 0.113***  |
|               | [3.89]       | [1.41]   | [1.13]   | [3.90]     | [16.33]   | [10.26]   |
| <b>LTD</b>    | 0.1775695**  | 0.0124   | -0.0189  | -0.0167    | -0.195*** | -0.0953   |
|               | [2.23]       | [0.28]   | [-0.34]  | [0.32]     | [-2.65]   | [-1.03]   |
| <b>BI</b>     | 0.2647368*   | 0.0506   | -0.0716  | -0.179     | -0.0846   | -0.199**  |
|               | [1.74]       | [0.86]   | [-0.99]  | [-1.47]    | [-0.84]   | [-2.50]   |
| <b>GDP</b>    | 0.4732932**  | 0.0679   | 0.246    | 0.164      | 0.469     | 0.459     |
|               | [2.45]       | [0.24]   | [0.79]   | [0.62]     | [0.88]    | [0.76]    |
| <b>INF</b>    | -2.159927*** | 0.0640   | -0.104   | 0.0480     | 0.169     | 0.333     |
|               | [-3.63]      | [0.48]   | [-0.57]  | [0.18]     | [0.63]    | [1.06]    |
| <b>_cons</b>  | 1.822373**   | 0.745*** | 0.978*** | 1.259***   | 1.218***  | 1.586***  |

|                                       |        |        |        |        |        |        |
|---------------------------------------|--------|--------|--------|--------|--------|--------|
|                                       | [2.77] | [3.36] | [4.01] | [5.64] | [4.55] | [6.25] |
| <b>N</b>                              | 214    | 378    | 378    | 378    | 378    | 378    |
| <i>Note: t statistics in brackets</i> |        |        |        |        |        |        |
| *p<0.1, ** p<0.05, *** p<0.01         |        |        |        |        |        |        |

*Source: Compiled by the author using Stata 17 software.*

### **Bank Size (lnSIZE)**

Referring to Table 4.13, the SGMM results and most quantiles indicate that bank size has a statistically significant impact on financial sustainability (FSR). Specifically, the SGMM regression coefficient for lnSIZE is -0.071 at a 1% significance level. This implies that a one-unit decrease in bank size corresponds to a 0.071-unit increase FSR at a 1% significance level. This finding suggests that smaller banks tend to have lower financial collapse risk, thereby enhancing financial sustainability. This could be attributed to the fact that smaller banks often adopt more cautious risk management strategies, maintain higher capital buffers, and engage less in high-risk investment activities compared to larger banks.

Furthermore, to provide a more comprehensive and detailed analysis, the study examines the impact of bank size on financial sustainability across different quantiles. The quantile regression results at the 10%, 25%, 50%, 75%, and 90% consistently show a negative relationship Between bank size and FSR. This indicates that larger banks generally exhibit lower financial sustainability. This finding supports the hypothesis that larger banks may face higher risks due to the complexity of their operations, higher administrative costs, and easier access to external financing, which may encourage greater risk-taking behavior.

Notably, the negative impact of lnSIZE on FSR intensifies at higher quantiles, increasing from -0.030 at the 10 percentile to -0.128 at the 90 percentile. This suggests that for banks with higher financial sustainability, the adverse effects of bank size become more pronounced. In other words, among banks with strong financial stability, expanding their size may be associated with greater risks, ultimately

undermining financial sustainability. This result is consistent with the findings of Daoud and Kammoun (2020), which argue that larger size does not necessarily equate to greater financial stability but rather depends on risk management capabilities and operational efficiency.

**Therefore, the hypothesis H1:** *Bank size (lnSIZE) has a negative impact on the financial sustainability of Vietnamese commercial banks* is accepted.

### **Return on Assets (ROA)**

Referring to Table 4.13, the SGMM results and all quantile estimates indicate that the impact of Return on Assets (ROA) on financial sustainability (FSR) is statistically significant. Specifically, the SGMM regression coefficient for ROA is 5.359 at a 1% significance level, implying that a one-unit increase in ROA leads to a 5.359-unit increase in FSR. ROA reflects the efficiency of a bank in utilizing its assets to generate profit. A higher ROA signifies strong profitability and effective asset utilization, whereas a lower ROA suggests difficulties in generating substantial profits. A higher ROA provides banks with sufficient financial resources to withstand financial risks, thereby enhancing financial sustainability.

To gain a more detailed and comprehensive understanding, the study further analyzes the impact of ROA on financial stability across different quantiles to draw more precise conclusions. The quantile regression results at 10%, 25%, 50%, 75%, and 90% (as presented in Table 4.13) indicate that ROA maintains a positive and statistically significant coefficient across most quantiles, with the strongest effects observed at Q25 (12.69) and Q50 (11.16). This suggests that banks with moderate financial sustainability levels benefit the most from improved profitability. Conversely, at Q75 (9.81) and Q90 (10.88), the positive impact of ROA on FSR remains but slightly weakens, indicating that for banks with high financial sustainability, profitability remains important but is no longer the sole determining factor, as other elements also come into play.

This finding aligns with theoretical expectations, suggesting that higher profitability provides banks with more financial resources to manage risks and maintain stability. Moreover, it is consistent with the research of Meher & Getaneh (2019) and Moechdi et al. (2016), which emphasize that banks with higher profitability are generally more resilient to financial shocks.

**Therefore, the hypothesis H2:** *Return on Assets (ROA) has a positive impact on the financial sustainability of Vietnamese commercial banks* is accepted.

### **Capital Adequacy Ratio (CAR)**

According to the SGMM regression results, the coefficient of CAR (Capital Adequacy Ratio) is -0.1048, but it is not statistically significant. This means that there is insufficient evidence and reliability to conclude whether CAR has a positive or negative impact on the financial sustainability of Vietnamese commercial banks.

When analyzing CAR across different quantiles, significant variations in coefficients are observed. Specifically, at the 75th quantile (Q75), CAR has a positive and statistically significant impact at the 5% level, with a coefficient of 0.391. This suggests that an increase in capital adequacy ratio enhances risk resilience and strengthens financial stability, thereby contributing to a higher level of financial sustainability for Vietnamese commercial banks.

Conversely, at lower quantiles (Q10, Q25, Q50), the coefficient of CAR is not statistically significant, indicating that for banks with low or average financial sustainability, capital adequacy does not appear to be a determining factor in their financial stability. This finding partially aligns with the research of Berger & Bouwman (2013), which suggests that the impact of capital on bank stability depends on individual bank characteristics and risk tolerance levels. However, overall, the regression coefficients across other quantiles remain insignificant, meaning there is no strong basis for a conclusive finding. This differs from the initial expectations of the study.

**Therefore, the hypothesis H3:** *Capital Adequacy Ratio (CAR) has a positive impact on the financial sustainability of Vietnamese commercial banks* is accepted (at Q75).

#### **Non-Performing Loans (NPL)**

Examining the regression results, the SGMM coefficient for NPL is -0.979 and is statistically significant at the 10% level, indicating that an increase of 1 unit in the NPL ratio leads to a decrease of 0.979 units in the FSR index. This suggests that as the non-performing loan (NPL) ratio rises, banks face higher risks and become more financially vulnerable, ultimately reducing their financial sustainability. NPL represents the portion of outstanding loans that banks struggle to recover, or in the worst case, may be entirely unrecoverable. A higher NPL ratio can trigger negative financial consequences, affecting not only individual banks but also the overall banking system. However, some banks may turn NPL challenges into opportunities for development and stability, provided they can effectively manage rising bad debts.

However, when analyzing the quantile regression results (Q10, Q25, Q50, Q75, Q90), NPL is not statistically significant at any quantile, despite all coefficients being negative. This implies that while NPLs may negatively impact financial sustainability, the effect is not statistically strong across specific quantile groups. In other words, the extent of NPL's influence on financial sustainability varies among different types of banks.

**Therefore, the hypothesis H4:** *Non-Performing Loans (NPL) have a negative impact on the financial sustainability of Vietnamese commercial banks* is not accepted.

#### **Liquidity Ratio (LIQ)**

The regression results show that the SGMM coefficient for LIQ is -0.940 and not statistically significant, indicating that liquidity does not have a clear impact on financial sustainability when considering the entire sample. One possible explanation is that large banks often have well-diversified investment portfolios and credit

lending strategies managed by experienced professionals, making liquidity fluctuations less impactful. Additionally, these banks benefit from higher credibility and stronger support from the central bank, smaller banks, and even government-backed guarantees, allowing them to mobilize substantial liquidity to absorb financial risks. In contrast, smaller banks, with limited resources and human capital, are more vulnerable to liquidity fluctuations and find it harder to meet liquidity requirements effectively.

However, when examining different quantiles, the liquidity ratio shows a statistically significant positive effect across all quantiles (Q10, Q25, Q50, Q75, Q90), with coefficients ranging from 0.385 to 0.705. This suggests that improving liquidity can significantly enhance financial sustainability. Banks with higher liquidity levels are better equipped to handle short-term obligations and mitigate the adverse effects of unexpected financial shocks, especially in an environment where market fluctuations, credit risks, and economic uncertainty place increasing pressure on the banking system.

**As a result, the hypothesis H5:** *Liquidity ratio (LIQ) has a positive impact on the financial sustainability of Vietnamese commercial banks* is accepted.

### **Operating Income (lnINC)**

The SGMM coefficient for lnINC is 0.065 and statistically significant at the 1% level, indicating a positive impact of operating income on financial sustainability. Specifically, a 1-unit increase in operating income raises the Financial Sustainability Ratio (FSR) by 0.065 units. This suggests that banks with higher income levels tend to have greater financial sustainability, as stronger financial resources enable them to better withstand risks.

The positive effect of operating income is not only evident in the SGMM regression results but also consistently observed across all quantiles in the quantile regression model, with coefficients ranging from 0.0227 (Q25) to 0.113 (Q90). Notably, this effect remains statistically significant at the 1% level for mid-to-high quantiles (Q50, Q75, Q90), reinforcing the critical role of operating income in maintaining financial sustainability. As banks generate higher revenues, they have more resources to strengthen capital buffers, enhance liquidity, expand safer investment portfolios, and mitigate risks associated with non-performing loans (NPLs).

This finding aligns with modern banking theories, which emphasize that stable and diversified operating income enhances a bank's resilience to financial shocks. The study by Nguyen Thi Thuy (2023), which argues that operating income serves as a financial cushion, allowing banks to manage market risks more effectively while improving operational efficiency and risk management capabilities. Additionally, the relationship between income and financial sustainability can be explained through adaptive business strategies—banks with stable profitability can invest in technology, service quality, revenue diversification, and market competitiveness, thereby strengthening their position and minimizing liquidity risks over the long term.

Overall, these findings underscore the critical role of operating income in ensuring the financial sustainability of banks. This suggests that banks should diversify their revenue streams, moving beyond traditional interest income to fee-based services, financial investments, and other business activities. At the same time, cost control and risk management remain key to maintaining financial stability and sustainable growth amid an increasingly volatile market environment.

**As a result, the hypothesis H6:** *Operating income (lnINC) has a positive impact on the financial sustainability of Vietnamese commercial banks* is accepted.

### **Loan to Deposit Ratio (LTD)**

The SGMM regression results show that the coefficient of LTD is 0.177, with statistical significance at the 5% level. This indicates that a 1-unit increase in the Loan to Deposit Ratio (LTD) raises the Financial Sustainability Ratio (FSR) by 0.177 units. The findings suggest that a higher LTD ratio contributes to improved financial sustainability. Specifically, banks with higher LTD ratios can optimize profits from lending activities, thereby enhancing profitability and strengthening financial sustainability. A well-managed expansion of credit allows banks to efficiently utilize mobilized funds, generate stable income from interest rate spreads, and maintain a steady cash flow to support operational needs.

However, the quantile regression analysis reveals notable variations across different bank groups. At the Q75 quantile, the coefficient of LTD is -0.195 and statistically significant at the 1% level, suggesting that for banks with higher financial sustainability, excessive credit expansion may lead to increased risks. When the LTD ratio surpasses a safe threshold, banks may face rising non-performing loans (NPLs), deteriorating asset quality, and adverse effects on financial sustainability. This reflects the risk that over-reliance on lending activities without stringent risk management can make banks more vulnerable to economic fluctuations and credit risks.

Conversely, at lower quantiles (Q10, Q25, Q50), the coefficient of LTD is not statistically significant, indicating that for banks with moderate or lower financial sustainability, LTD is not a key determinant. This can be explained by the fact that these banks often adopt more conservative credit management strategies, focusing on liquidity maintenance and credit risk mitigation rather than maximizing lending profitability.

Overall, the SGMM findings align with previous studies on the relationship between LTD and bank financial sustainability. Research suggests that maintaining an optimal LTD ratio allows banks to maximize profitability without excessively

increasing credit risk. However, exceeding a safe LTD threshold can heighten liquidity risks, ultimately weakening financial sustainability.

**Therefore, the hypothesis H7:** *The Loan to Deposit Ratio (LTD) has a positive impact on the financial sustainability of Vietnamese commercial banks* is not accepted.

### **Board Independence Ratio (BI)**

The results in Table 4.13 indicate that the Board Independence Ratio (BI) has a positive impact on Financial Sustainability (FSR). The SGMM regression coefficient for BI is 0.265, with statistical significance at the 10% level, implying that a 1-unit increase in the proportion of independent board members leads to a 0.265-unit increase in FSR. This suggests that a higher proportion of independent board members can enhance transparency and improve risk control, thereby strengthening financial sustainability. Since independent board members are not bound by personal or internal bank interests, they can objectively monitor executive management and ensure better corporate governance.

However, the quantile regression analysis reveals significant variations in BI's impact across different bank groups. At lower quantiles (Q10, Q25, Q50, Q75), the BI coefficient is not statistically significant, indicating that for banks with low to moderate financial sustainability, the proportion of independent board members is not a key determinant. In contrast, at Q90, the BI coefficient is -0.199, statistically significant at the 5% level, suggesting that a higher proportion of independent board members may actually reduce financial sustainability. This could be due to the fact that an excessively high number of independent board members may slow down decision-making processes or misalign strategic direction, negatively affecting operational efficiency.

Overall, an optimal proportion of independent board members can improve bank governance, an excessive proportion may reduce flexibility and hinder effective decision-making.

**Therefore, the hypothesis H8:** *The Board Independence Ratio (BI) has a positive impact on the financial sustainability of Vietnamese commercial banks* is accepted.

### **Economic Growth (GDP)**

The SGMM regression results indicate that the coefficient of GDP is 0.473, statistically significant at the 5% level, suggesting that economic growth positively contributes to the financial sustainability (FSR) of banks. This finding aligns with expectations, as higher economic growth is generally associated with increased income levels, reduced non-performing loan (NPL) ratios, and improved bank profitability. The results are consistent with the studies of Gupta & Kashiramka (2020) and Rupeika-Apoga et al. (2018).

However, when analyzing the impact across different quantiles, the GDP coefficient is not statistically significant, indicating that the effect of economic growth may not be uniform across different bank groups. One possible explanation is that larger and more stable banks may benefit more from economic expansion, while smaller or high-risk banks may not experience the same positive impact. Additionally, rapid economic growth may sometimes be accompanied by an increase in risky credit expansion, especially if banks overextend lending without adequate credit quality control. The overall effect of economic growth on banking stability can vary depending on the market structure and financial policies of each country.

**Therefore, the hypothesis H9:** *Economic growth (GDP) has a positive impact on the financial sustainability of Vietnamese commercial banks* is not accepted.

### **Inflation rate (INF)**

The SGMM results show that the coefficient of INF is -2.159, statistically significant at the 1% level, indicating that inflation has a negative impact on financial

sustainability. This outcome is consistent with expectations, as higher inflation can erode asset values, increase funding costs, and weaken customer purchasing power, all of which negatively affect banking operations. Moreover, rising inflation reduces the real value of money, making it more difficult for individuals and businesses to repay loans with devalued currency. This, in turn, weakens debt repayment capacity and increases credit risk for banks.

However, when analyzing quantile regression results, the INF coefficient is not statistically significant, implying that the impact of inflation on financial stability is not uniform across different bank groups. Some banks may be better equipped to adapt to high-inflation environments, particularly those with strong risk management strategies and well-diversified asset portfolios.

This finding is in line with the research of Lan (2021) and Rahim et al. (2012), which suggests that high inflation is often associated with declining credit quality and increased informal credit growth, exacerbating risks in the banking system.

**Thus, the hypothesis H10:** *Inflation (INF) has a negative impact on the financial sustainability of Vietnamese commercial banks* is not accepted.

**Table 4. 14: Summary of Hypotheses and Empirical Research Results**

| <b>Variable</b>                                                | <b>Expected impact</b> | <b>Research results</b> | <b>Significance level</b>                   | <b>Hypothesis</b> |
|----------------------------------------------------------------|------------------------|-------------------------|---------------------------------------------|-------------------|
| <b>lnSIZE</b>                                                  | -                      | -                       | SGMM (1%)<br>Q50;Q75;Q90 (1%)               | Accepted H1       |
| <b>ROA</b>                                                     | +                      | +                       | SGMM;<br>Q25;Q50;Q75;Q90<br>(1%)<br>Q10(5%) | Accepted H2       |
| <b>CAR</b>                                                     | +                      | +                       | Q75 (1%)                                    | Accepted H3       |
| <b>NPL</b>                                                     | -                      |                         | Not statistically significant               | Rejected H4       |
| <b>LIQ</b>                                                     | +                      | +                       | Q10;Q25 (5%)<br>Q50;Q75 (1%)<br>Q90 (10%)   | Accepted H5       |
| <b>lnINC</b>                                                   | +                      | +                       | SGMM;Q50;Q75;Q90<br>(1%)                    | Accepted H6       |
| <b>LTD</b>                                                     | -                      | -                       | SGMM (5%)<br>Q75 (5%)                       | Accepted H7       |
| <b>BI</b>                                                      | +                      | +                       | SGMM;Q90 (10%)                              | Accepted H8       |
| <b>GDP</b>                                                     | +                      |                         | Not statistically significant               | Rejected H9       |
| <b>INF</b>                                                     | -                      |                         | Not statistically significant               | Rejected H10      |
| <p><b><u>Note:</u></b><br/>(+): Positive<br/>(-): Negative</p> |                        |                         |                                             |                   |

*Source: Compiled by the author using Stata 17 software.*

## **SUMMARY OF CHAPTER 4**

Chapter 4 provides a comprehensive analysis of the factors influencing the financial sustainability of Vietnamese commercial banks during the 2010–2023 period. The research findings indicate that return on assets (ROA), capital adequacy ratio (CAR), liquidity ratio (LIQ), operating income (lnINC) the proportion of independent board members (BI), have a positive impact on financial sustainability. In contrast, bank size (lnSIZE), loan-to-deposit ratio (LTD), exert a negative influence. Additionally, non-performing loans (NPL), economic growth (GDP) and inflation (INF) when analyzing quantile regression results, coefficient is not statistically significant

The application of the SGMM method and quantile analysis provides deeper insights into the variations across different groups of banks, offering critical empirical evidence for policymakers and bank managers to formulate appropriate strategies aimed at enhancing financial sustainability in Vietnam's commercial banking system.

## CHAPTER 5. CONCLUSIONS AND IMPLICATION

### 5.1. CONCLUSIONS

This study analyzes the factors influencing the financial sustainability of Vietnamese commercial banks during the 2010–2023 period. Using the System Generalized Method of Moments (SGMM) regression and quantile regression, the study examines the impact of microeconomic factors (bank-specific characteristics) and macroeconomic conditions (economic environment) on the Financial Sustainability Ratio (FSR). Panel data regression models, including Pooled OLS, Fixed Effects Model (FEM), and Random Effects Model (REM), were employed to estimate the factors affecting financial sustainability. After conducting necessary diagnostic tests, the REM was found to be more appropriate than Pooled OLS and FEM. However, issues of heteroskedasticity and autocorrelation persisted. To address these limitations, the Feasible Generalized Least Squares (FGLS) method was applied to improve estimation accuracy. Given the potential endogeneity of certain variables, the System Generalized Method of Moments (SGMM) was further employed to mitigate endogeneity bias and ensure the reliability of the findings. Additionally, quantile regression was used to gain deeper insights into the varying impacts of explanatory variables across different levels of financial sustainability.

The findings reveal that bank size ( $\ln\text{SIZE}$ ) negatively impacts financial sustainability, indicating that larger banks face higher risks due to expanded operations and increased complexity. Return on Assets (ROA) exhibits a positive effect, suggesting that banks with higher profitability have a stronger financial foundation and are better equipped to withstand unexpected risks. Meanwhile, the capital adequacy ratio (CAR) is statistically insignificant in the SGMM model but shows a positive effect at higher quantiles (Q75), implying that capital adequacy has a greater influence on banks with high financial sustainability levels. Non-Performing Loans (NPLs) generally have a negative effect on financial sustainability by increasing liquidity, profitability, and credit risks, potentially harming a bank's

stability and reputation. However, quantile regression results show that NPLs are not statistically significant at any quantile, despite all coefficients being negative. This suggests that while NPLs may adversely impact financial sustainability, their effect is not consistently strong across different bank types or levels of financial health. Additionally, liquidity (LIQ) positively impacts FSR across all quantiles, indicating that higher liquidity buffers help banks mitigate financial risks. Operating income (lnINC) has a significant positive effect on FSR, emphasizing the importance of income generation in strengthening banks' risk resilience. While the loan-to-deposit ratio (LTD) exhibits a positive effect in the SGMM model, indicating that moderate credit expansion can optimize funding utilization and enhance profitability, it has a negative impact at Q75, suggesting that excessive credit expansion may weaken financial sustainability. The findings further reveal that the proportion of independent board members (BI) has a positive effect on FSR, highlighting their critical role in risk governance, transparency, and mitigating conflicts of interest between management and shareholders. A higher proportion of independent directors leads to more objective decision-making and better oversight of credit and investment risks, thereby enhancing banks' financial sustainability. Regarding macroeconomic factors, the study finds that GDP growth positively impacts financial sustainability, as economic expansion improves credit quality, reduces NPLs, and enhances bank profitability. Conversely, inflation (INF) has a negative effect, as it raises funding costs, erodes the real value of collateral, and weakens borrowers' debt repayment capacity, thereby reducing financial sustainability. These empirical findings provide valuable insights into the role of both microeconomic and macroeconomic factors in maintaining the financial sustainability of Vietnamese commercial banks. Furthermore, the study offers policy implications to help banks enhance risk management, optimize business strategies, and ensure financial sustainability amid increasing market volatility. However, when analyzing the impact across different quantiles, the GDP and INF coefficient is not statistically significant.

## 5.2. IMPLICATIONS

The global economy is currently experiencing significant challenges and volatility. According to United Nations forecasts, global economic growth is projected to reach 2.7% this year. However, developed economies continue to face risks of slowing growth, despite the gradual easing of monetary policies. While the global economy has not entered a recession, it remains below its potential, with inflation levels declining compared to 2023 but still above target levels in many countries. Despite these moderating inflation trends, major economies continue to uphold tight monetary policies and maintain high interest rates. Meanwhile, global trade, consumption, and investment activity continue to weaken, exacerbated by rising protectionist measures and trade defense mechanisms, which are disrupting import-export activities at both regional and global levels. Notably, the financial crisis of Saigon Commercial Bank (SCB) in Vietnam in 2022 triggered by leadership instability and adverse rumors everely impacted market confidence in the banking sector. These events underscore the vulnerability of the financial system in the face of unexpected shocks, emphasizing the importance of robust risk management frameworks. Consequently, banks must proactively develop adaptive strategies to mitigate risks, ensure liquidity, and maintain financial sustainability amid an increasingly volatile global economic landscape.

Based on the research findings on the factors influencing the financial sustainability of Vietnamese commercial banks during the 2010–2023 period, the author proposes the following policy recommendations to enhance the financial sustainability of Vietnam's commercial banking sector:

Firstly, the research findings indicate that bank size ( $\ln\text{SIZE}$ ) has a negative impact on the Financial Sustainability Ratio (FSR), suggesting that an increase in scale does not necessarily enhance financial sustainability. While expanding scale provides banks with competitive advantages, excessive and uncontrolled growth can expose them to liquidity pressures and asset quality deterioration. Therefore, banks

should adopt a prudent expansion strategy, ensuring that governance capacity and risk management systems develop in alignment with the pace of scale growth to maintain financial stability and operational efficiency.

Secondly, the research findings indicate that Return on Assets (ROA) has a positive impact on the Financial Sustainability Ratio (FSR), suggesting that banks with higher profitability tend to have a stronger financial foundation. To optimize profitability, banks should enhance cost management efficiency, adopt modern technology to minimize operating expenses, and optimize investment portfolios to improve profitability. However, while ROA growth enhances profitability, it may also introduce financial risks, particularly liquidity risk, which can significantly affect a bank's stability. Therefore, increasing ROA must be carefully monitored and effectively managed. First, banks should optimize asset allocation by diversifying investment and loan portfolios into higher-yield assets while minimizing credit risk exposure. Additionally, they must establish rigorous credit assessment frameworks, conduct regular monitoring and evaluation of loans and investment portfolios, and ensure robust due diligence procedures. Furthermore, banks should reassess their cost structure to effectively manage operating expenses, allowing them to improve operational efficiency without compromising service quality. Additionally, banks can expand into low-risk, revenue-generating services such as wealth management, investment advisory, and M&A consulting, which contribute to sustainable profitability while mitigating excessive risk exposure.

Thirdly, the findings suggest that the Capital Adequacy Ratio (CAR) only has a positive and statistically significant impact on the financial sustainability of Vietnamese commercial banks at the quantile Q75, while it remains insignificant at lower and median quantiles. This implies that increasing CAR is effective in enhancing risk resilience and long-term stability primarily for financially strong banks. In contrast, for banks with low to average financial sustainability, CAR does not appear to be a key determinant. Therefore, these banks may need to focus more

on improving asset quality, operational efficiency, and risk management. The results indicate that capital-related policies should be tailored to individual bank characteristics rather than applied uniformly across the sector.

Fourth, The regression findings reveal that non-performing loans (NPLs) significantly undermine the financial sustainability of Vietnamese commercial banks at the overall level, as evidenced by the SGMM model. This highlights the importance of credit quality and effective loan recovery mechanisms in ensuring long-term stability. A higher NPL ratio erodes bank profitability, strains liquidity, and increases credit risk exposure, thereby weakening a bank's financial position. These findings underscore the need for stricter credit risk management policies and early warning systems to detect and address loan defaults. However, the absence of statistical significance across all quantiles in the quantile regression model indicates that the negative impact of NPLs is not consistent across banks with different levels of financial sustainability. This variation suggests that while some banks may be more vulnerable to rising NPLs, others may have sufficient risk absorption capacity or recovery mechanisms in place. Therefore, policymakers and bank managers should adopt a more tailored approach in managing bad debt risks, taking into account each bank's size, risk appetite, and operational resilience.

Fifth, the findings highlight that while Liquidity Ratio (LIQ) may not show a significant impact on financial sustainability across the full sample, its importance becomes evident when analyzed at different levels of bank performance. The significant positive effect of the liquidity ratio across all quantiles suggests that maintaining adequate liquidity is essential for banks to strengthen resilience against short-term shocks and financial stress. This is particularly crucial for smaller banks that lack the diversified portfolios and institutional support of larger institutions. Therefore, policymakers and bank managers should prioritize liquidity risk management, especially for less stable banks, by enhancing liquidity buffers and

implementing robust monitoring mechanisms to improve overall financial sustainability in the commercial banks of Vietnamese

Sixth, the research findings indicate that operating income (lnINC) has a positive impact on a bank's financial sustainability (FSR). This emphasizes that increasing operating income not only improves business efficiency but also enhances financial resilience, enabling banks to better withstand economic shocks and mitigate risks associated with fluctuations in interest rates and credit markets. A stable and growing stream of operating income serves as a crucial factor in ensuring long-term financial sustainability, as it strengthens capital buffers, supports investment in technology and infrastructure, and enhances overall institutional stability. To capitalize on this, banks should focus on diversifying revenue sources, reducing over-reliance on interest income, and expanding into non-interest financial services such as electronic payments, asset management, insurance, and investment advisory. Developing a well-balanced revenue structure will help banks maintain steady earnings, even during periods of economic downturn or credit market instability. Additionally, leveraging digital transformation and artificial intelligence in banking operations will enable institutions to optimize operational costs, expand market reach, improve service efficiency, and enhance customer experience. Furthermore, the integration of big data analytics and machine learning into financial services can improve risk assessment, fraud detection, and personalized banking solutions, ultimately driving sustainable profitability. By proactively adapting to technological advancements and evolving customer preferences, banks can establish a more resilient and adaptive business model, ensuring both short-term performance improvements and long-term financial sustainability.

Seventh, the Loan-to-Deposit Ratio (LTD) has a positive impact on the Financial Sustainability Ratio (FSR) in the SGMM model; however, at higher quantiles, it exhibits a negative effect. This suggests that while credit expansion can contribute positively to financial sustainability if properly managed, excessive

lending beyond a safe threshold may pose significant risks. Therefore, banks must strike a balance between leveraging credit growth opportunities and maintaining prudent risk controls, particularly in terms of liquidity risk and non-performing loans (NPLs). Strengthening credit portfolio monitoring is essential to ensure that loan expansion does not lead to an increase in NPLs (non-performing loans), which could erode financial stability. Additionally, banks should focus on diversifying funding sources beyond traditional customer deposits by exploring alternative financing channels such as bond issuance, interbank borrowing, and international financial partnerships. To enhance risk management, banks should adopt advanced credit risk management frameworks and integrate artificial intelligence (AI)-driven predictive models to forecast liquidity risks associated with excessive LTD levels. By leveraging AI and data analytics, banks can improve liquidity planning, stress testing, and early-warning mechanisms, ensuring a more resilient and sustainable credit growth strategy.

Eighth, the research findings indicate that the proportion of independent board members (BI) has a positive impact on financial sustainability; however, at higher quantiles, it shows signs of a negative effect. This suggests that while independent board members play a crucial role in overseeing banking operations, an excessively high proportion may reduce decision-making flexibility. Therefore, banks should maintain a balanced ratio between independent and executive board members, ensuring that their presence is not merely for regulatory compliance but also for substantive contributions to governance and strategic planning. The optimal ratio should be determined based on the bank's size and the complexity of its operations. Moreover, to ensure that independent board members effectively contribute to oversight and strategy formulation, banks should prioritize individuals with deep expertise in finance, risk management, and banking operations. Appointing independent solely to meet regulatory requirements should be avoided; instead, banks should focus on selecting qualified professionals who can provide valuable insights and guidance. Additionally, banks need to establish an effective

collaboration framework between independent board members and the executive management team, ensuring seamless coordination in decision-making processes. The board of directors should maintain flexibility in adjusting its composition to align with the bank's strategic development objectives at different stages. Additionally, banks must implement transparent oversight mechanisms and establish robust internal control systems to enable independent board members to effectively fulfill their supervisory roles.

Finally, The empirical results highlight that macroeconomic conditions, particularly economic growth and inflation, play influential—but uneven—roles in shaping the financial sustainability of Vietnamese commercial banks. The SGMM regression confirms that GDP growth positively affects financial sustainability, supporting the idea that a thriving economy fosters higher income levels, improved loan performance, and stronger profitability. However, the lack of significance across quantiles suggests this positive impact is not evenly distributed—larger or more resilient banks may benefit more, while smaller or riskier banks may not realize the same advantages. Policymakers should, therefore, focus on inclusive growth strategies and ensure that smaller institutions are also positioned to capitalize on macroeconomic improvements.

Conversely, inflation shows a significantly negative effect on financial sustainability at the full-sample level, reflecting how rising prices increase operational costs, erode asset values, and deteriorate customers' repayment capacity. Still, its non-significance across different quantiles reveals that not all banks are equally vulnerable. Banks with robust capital buffers, diversified portfolios, or effective risk management frameworks may be better shielded from inflationary pressures. These findings imply that regulatory authorities and bank managers should prioritize inflation risk mitigation—especially through prudent lending standards, stress testing, and adaptive financial planning—to strengthen resilience across the

banking sector. Ultimately, while macroeconomic conditions matter, internal institutional strength remains key to sustaining financial health amid economic volatility.

### 5.3. LIMITATIONS

The study on The factors influencing the financial sustainability of Vietnamese Commercial Banks: A quantile regression analysis has yielded significant findings. However, several limitations must be acknowledged to enhance the study's reliability and comprehensiveness:

Firstly, the study is constrained by a relatively short research timeframe. The three-month duration allocated for the thesis is insufficient to conduct a comprehensive and in-depth study. The tight schedule has created challenges in data collection, processing, and model validation, limiting the ability to ensure full accuracy and reliability. Certain key steps, such as testing advanced analytical models or assessing the impact of capital structure under different economic conditions (e.g., during crises or recovery periods), could not be fully implemented.

Secondly, the study has significant spatial and temporal limitations. Specifically, the study uses data from 27 Vietnamese commercial banks, without extending the analysis to foreign banks or smaller financial institutions, such as cooperative banks or non-bank financial institutions. As a result, the findings may not fully capture the characteristics of the entire financial system. From a temporal perspective, the dataset covers the period 2010–2023, which may not comprehensively reflect the long-term effects of economic cycles. Additionally, the study does not explicitly analyze periods of financial crises or major disruptions in the banking sector, which could affect the generalizability of the results. Notably, the research primarily relies on financial statements from banks and macroeconomic data published by official sources. However, the quality of financial reporting may be influenced by variations in accounting regulations across banks or potential discrepancies in financial disclosures, which could impact the reliability of the data.

Thirdly, the study has methodological limitations. Although the SGMM model effectively addresses endogeneity issues, it may still face challenges in selecting instrumental variables, which could affect the accuracy of parameter estimates. Other

econometric approaches, such as Bayesian Estimation, Machine Learning, or Vector Auto Regression (VAR), could provide a deeper examination of the relationships between variables. Moreover, the study assumes that the factors affecting financial sustainability follow a linear relationship. However, these relationships may also exhibit nonlinear characteristics, meaning that the degree of impact could vary across different development stages of banks or under different economic conditions.

Fourth, the study has limitations in the selection of research variables. The research primarily focuses on internal banking factors, such as bank size, non-performing loans (NPLs), liquidity, and operating income, without considering other crucial elements that could influence financial sustainability. These include risk management quality, financial technology (Fintech) adoption, digital transformation levels, and international integration. Given the increasing importance of these factors in the modern banking landscape, their absence may limit the study's comprehensiveness.

Finally, the findings may be specific to the Vietnamese commercial banking sector. Although the study provides valuable insights into Vietnam's banking system, its findings may not necessarily be applicable to banks in other countries, particularly those in developed financial markets. Certain conclusions may be influenced by the unique characteristics of Vietnam's financial system, such as strong regulatory intervention by the State Bank of Vietnam (SBV) and specific credit policies, which are not necessarily present in other economies.

#### **5.4. FURTHER RESEARCH DIRECTIONS**

In the future, research can be expanded in various directions to overcome current limitations and provide deeper insights into the financial sustainability of banks.

Firstly, the data scope can be extended both temporally and spatially, covering a longer time horizon to analyze the cyclical effects of macroeconomic factors, while also incorporating comparisons with banks in the region to assess the generalizability of the findings.

Additionally, testing advanced analytical models, such as nonlinear models, artificial neural networks (Neural Networks), or Machine Learning techniques, can help examine the complex relationships between variables and improve the accuracy of financial sustainability forecasts.

Moreover, future studies can incorporate additional factors such as the degree of banking digitalization, the impact of Fintech, risk management quality, and the influence of regulatory frameworks like Basel III on banking stability.

Furthermore, analyzing the impact of these factors in the context of economic shocks, such as financial crises or interest rate fluctuations, will provide banks with valuable insights on how to respond and adapt to changing economic environments.

Finally, future research can focus on practical policy measures that banks can implement to enhance financial sustainability, thereby offering specific strategic recommendations for policymakers and banking regulators.

## **SUMMARY OF CHAPTER 5**

Chapter 5 has synthesized the key result of the study. Based on these results, the research proposes practical policy recommendations to help banks optimize operations, enhance risk management efficiency, and adapt to economic fluctuations. Specifically, banks need to effectively control credit growth, maintain adequate liquidity, improve credit quality, and optimize investment portfolios to strengthen their risk resilience. Additionally, establishing an appropriate governance structure, leveraging financial technology, and adjusting strategies in response to economic cycles will support banks in achieving long-term sustainable growth.

However, the study has certain limitations related to data scope, analytical methods, research models, and unexamined factors. Therefore, future research can be expanded by collecting data over a longer period, testing advanced analytical models, and incorporating additional factors such as banking digitalization, the impact of Fintech, and regulatory policies from the State Bank of Vietnam. These research directions will provide a more comprehensive understanding of banking financial sustainability, while offering more precise and effective recommendations for the banking system amid an increasingly dynamic economic environment.

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## APPENDIX

### Appendix 1: Research data

|    | Stock code | Full name                                                               |
|----|------------|-------------------------------------------------------------------------|
| 1  | ABB        | An Binh Commercial Joint Stock Bank Viet Nam.                           |
| 2  | ACB        | Asia Commercial Joint Stock Bank.                                       |
| 3  | AGRB       | Vietnam Bank for Agriculture and Rural Development.                     |
| 4  | BAB        | Bac A Commercial Joint Stock Bank.                                      |
| 5  | BID        | Joint stock Commercial Bank for Investment and Development of Viet Nam. |
| 6  | BVB        | Viet Capital Commercial Joint Stock Bank.                               |
| 7  | CTG        | Vietnam Joint Stock Commercial Bank for Industry and Trade.             |
| 8  | EIB        | Vietnam Export Import Commercial Joint Stock Bank                       |
| 9  | HDB        | Ho Chi Minh City Development Joint Stock Commercial Bank                |
| 10 | KLB        | Kien Long Commercial Joint Stock Bank.                                  |
| 11 | LPB        | Fortune Vietnam Joint Stock Commercial Bank                             |
| 12 | MBB        | Military Commercial Joint Stock Bank.                                   |
| 13 | MSB        | Vietnam Maritime Commercial Joint Stock Bank.                           |
| 14 | NAB        | Nam A Bank Commercial Joint Stock Bank.                                 |
| 15 | NVB        | National Citizen Commercial Joint Stock Bank.                           |
| 16 | OCB        | Orient Commercial Joint Stock Bank.                                     |
| 17 | PGB        | Prosperity and Growth Commercial Joint Stock Bank.                      |
| 18 | SGB        | Saigon Bank For Industry And Trade                                      |
| 19 | SHB        | Saigon – Hanoi Commercial Joint Stock Bank.                             |
| 20 | SSB        | Southeast Asia Commercial Joint Stock Bank.                             |
| 21 | STB        | Saigon Thuong Tin Commercial Joint Stock Bank.                          |
| 22 | TCB        | Vietnam Technological And Commercial Joint Stock Bank.                  |
| 23 | TPB        | Tien Phong Commercial Joint Stock Bank.                                 |
| 24 | VAB        | Vietnam Asia Commercial Joint Stock Bank.                               |
| 25 | VCB        | Bank for Foreign Trade of Vietnam                                       |
| 26 | VIB        | Vietnam International Commercial Joint Stock Bank.                      |
| 27 | VPB        | Vietnam Prosperity Joint Stock Commercial Bank.                         |

---

**Appendix 2: Descriptive statistical results**

```
. sum FSR lnSIZE ROA CAR NPL LIQ lnINC LTD BI GDP INF
```

| Variable | Obs | Mean     | Std. dev. | Min       | Max      |
|----------|-----|----------|-----------|-----------|----------|
| FSR      | 378 | 1.198396 | .182104   | .8203008  | 2.015621 |
| lnSIZE   | 378 | 18.75759 | 1.247001  | 15.92274  | 21.55653 |
| ROA      | 378 | .0092515 | .0088486  | -.0551175 | .09987   |
| CAR      | 378 | .1337347 | .0414289  | .0835     | .3342    |
| NPL      | 378 | .0232843 | .0283381  | .0001832  | .4029098 |
| LIQ      | 378 | .92605   | .0463757  | .7538871  | .9918574 |
| lnINC    | 378 | 14.31074 | 1.766321  | 10.26594  | 18.59307 |
| LTD      | 378 | .8711691 | .1814473  | .0236     | 1.55997  |
| BI       | 378 | .1395911 | .0824381  | 0         | .4       |
| GDP      | 378 | .06015   | .0154624  | .026      | .08      |
| INF      | 378 | .0516    | .0446574  | .006      | .1868    |

**Appendix 3: Correlation analysis**

```
. corr FSR lnSIZE ROA CAR NPL LIQ lnINC LTD BI GDP INF
(obs=378)
```

|        | FSR     | lnSIZE  | ROA     | CAR     | NPL     | LIQ     | lnINC   | LTD     |
|--------|---------|---------|---------|---------|---------|---------|---------|---------|
| FSR    | 1.0000  |         |         |         |         |         |         |         |
| lnSIZE | 0.1494  | 1.0000  |         |         |         |         |         |         |
| ROA    | 0.5433  | 0.2575  | 1.0000  |         |         |         |         |         |
| CAR    | 0.0622  | -0.5850 | 0.0161  | 1.0000  |         |         |         |         |
| NPL    | -0.1075 | -0.1028 | -0.1318 | 0.0317  | 1.0000  |         |         |         |
| LIQ    | 0.3046  | 0.4465  | 0.2298  | -0.1585 | -0.1271 | 1.0000  |         |         |
| lnINC  | 0.4780  | 0.8315  | 0.3604  | -0.4102 | -0.0641 | 0.4416  | 1.0000  |         |
| LTD    | 0.1957  | 0.2758  | 0.2875  | -0.0403 | -0.0353 | 0.3862  | 0.3028  | 1.0000  |
| BI     | -0.1702 | -0.0987 | -0.0444 | -0.0114 | 0.0302  | -0.0475 | -0.1424 | 0.1171  |
| GDP    | -0.0225 | -0.0947 | -0.0882 | 0.0915  | 0.0486  | -0.0337 | -0.0952 | -0.0969 |
| INF    | 0.0896  | -0.2894 | 0.0240  | 0.2840  | 0.0650  | -0.2802 | -0.1673 | -0.1537 |
|        | BI      | GDP     | INF     |         |         |         |         |         |
| BI     | 1.0000  |         |         |         |         |         |         |         |
| GDP    | -0.0460 | 1.0000  |         |         |         |         |         |         |
| INF    | -0.2712 | 0.0394  | 1.0000  |         |         |         |         |         |

F test that all  $u_i=0$ :  $F(28, 339) = 5.16$  Prob > F = 0.0000

## Appendix 6: REM regression results

```
. xtreg FSR lnSIZE ROA CAR NPL LIQ lnINC LTD BI GDP INF, re
```

|                               |                  |   |        |
|-------------------------------|------------------|---|--------|
| Random-effects GLS regression | Number of obs    | = | 378    |
| Group variable: MACK1         | Number of groups | = | 29     |
| R-squared:                    | Obs per group:   |   |        |
| Within = 0.5777               | min =            |   | 1      |
| Between = 0.6056              | avg =            |   | 13.0   |
| Overall = 0.5907              | max =            |   | 14     |
| corr(u_i, X) = 0 (assumed)    | Wald chi2(10)    | = | 505.69 |
|                               | Prob > chi2      | = | 0.0000 |

| FSR     | Coefficient | Std. err.                         | z     | P> z  | [95% conf. interval]   |
|---------|-------------|-----------------------------------|-------|-------|------------------------|
| lnSIZE  | -.1243656   | .0126663                          | -9.82 | 0.000 | -.1491912    -.09954   |
| ROA     | 7.55227     | .7223775                          | 10.45 | 0.000 | 6.136436    8.968104   |
| CAR     | -.0156791   | .1898069                          | -0.08 | 0.934 | -.3876938    .3563356  |
| NPL     | -.3459586   | .2011178                          | -1.72 | 0.085 | -.7401423    .0482251  |
| LIQ     | .5470449    | .1739163                          | 3.15  | 0.002 | .2061751    .8879147   |
| lnINC   | .1051727    | .0073728                          | 14.27 | 0.000 | .0907224    .119623    |
| LTD     | .0224171    | .0384142                          | 0.58  | 0.560 | -.0528733    .0977075  |
| BI      | -.2162099   | .0828394                          | -2.61 | 0.009 | -.3785721    -.0538477 |
| GDP     | .3563506    | .3488196                          | 1.02  | 0.307 | -.3273233    1.040025  |
| INF     | .0916323    | .1461889                          | 0.63  | 0.531 | -.1948926    .3781573  |
| _cons   | 1.444211    | .2276427                          | 6.34  | 0.000 | .9980396    1.890383   |
| sigma_u | .05793556   |                                   |       |       |                        |
| sigma_e | .10245406   |                                   |       |       |                        |
| rho     | .24228981   | (fraction of variance due to u_i) |       |       |                        |

## Appendix 7: Breusch-Pagan test results

```
. xttest0
```

Breusch and Pagan Lagrangian multiplier test for random effects

$$\text{FSR}[\text{MACK1}, t] = Xb + u[\text{MACK1}] + e[\text{MACK1}, t]$$

Estimated results:

|     | Var      | SD = sqrt(Var) |
|-----|----------|----------------|
| FSR | .0331619 | .182104        |
| e   | .0104968 | .1024541       |
| u   | .0033565 | .0579356       |

Test: Var(u) = 0

chibar2(01) = 118.81  
 Prob > chibar2 = 0.0000

## Appendix 8: Hausman test results

```
. hausman fem rem
```

|        | —— Coefficients —— |            | (b-B)<br>Difference | sqrt(diag(V_b-V_B))<br>Std. err. |
|--------|--------------------|------------|---------------------|----------------------------------|
|        | (b)<br>fem         | (B)<br>rem |                     |                                  |
| lnSIZE | -.1243101          | -.1243656  | .0000555            | .0105164                         |
| ROA    | 7.40222            | 7.55227    | -.1500503           | .1826761                         |
| CAR    | .0297344           | -.0156791  | .0454135            | .0637637                         |
| NPL    | -.3309922          | -.3459586  | .0149664            | .0377167                         |
| LIQ    | .5336593           | .5470449   | -.0133856           | .0700851                         |
| lnINC  | .1099211           | .1051727   | .0047484            | .0033977                         |
| LTD    | .033907            | .0224171   | .0114899            | .0157727                         |
| BI     | -.2468979          | -.2162099  | -.030688            | .0360657                         |
| GDP    | .3780952           | .3563506   | .0217446            | .0403423                         |
| INF    | .0900739           | .0916323   | -.0015585           | .0586979                         |

b = Consistent under  $H_0$  and  $H_a$ ; obtained from `xtreg`.

B = Inconsistent under  $H_a$ , efficient under  $H_0$ ; obtained from `xtreg`.

Test of  $H_0$ : Difference in coefficients not systematic

$$\begin{aligned}\chi^2(10) &= (b-B)'[(V_b-V_B)^{-1}](b-B) \\ &= \mathbf{8.63}\end{aligned}$$

Prob >  $\chi^2 = \mathbf{0.5678}$

( $V_b-V_B$  is not positive definite)

## Appendix 9: Heteroskedasticity test results

```
. xttest0
```

Breusch and Pagan Lagrangian multiplier test for random effects

$$FSR[MACK1,t] = Xb + u[MACK1] + e[MACK1,t]$$

Estimated results:

|     | Var      | SD = sqrt(Var) |
|-----|----------|----------------|
| FSR | .0331619 | .182104        |
| e   | .0104968 | .1024541       |
| u   | .0033565 | .0579356       |

Test:  $\text{Var}(u) = 0$

$$\begin{aligned}\text{chibar2}(01) &= \mathbf{118.81} \\ \text{Prob} > \text{chibar2} &= \mathbf{0.0000}\end{aligned}$$

**Appendix 10: Autocorrelation test**

```
. xtserial FSR lnSIZE ROA CAR NPL LIQ lnINC LTD BI GDP INF
```

Wooldridge test for autocorrelation in panel data

H0: no first-order autocorrelation

F( 1, 26) = 49.759

Prob > F = 0.0000

**Appendix 11: Multicollinearity test result**

```
. vif
```

| Variable | VIF  | 1/VIF    |
|----------|------|----------|
| lnSIZE   | 4.53 | 0.220662 |
| lnINC    | 3.69 | 0.270646 |
| CAR      | 1.69 | 0.591628 |
| LIQ      | 1.49 | 0.673075 |
| INF      | 1.32 | 0.759000 |
| LTD      | 1.31 | 0.765193 |
| ROA      | 1.27 | 0.788361 |
| BI       | 1.17 | 0.856657 |
| NPL      | 1.04 | 0.960373 |
| GDP      | 1.03 | 0.973980 |
| Mean VIF | 1.85 |          |

**Appendix 12: Estimation results of the FGLS model**

Cross-sectional time-series FGLS regression

Coefficients: **generalized least squares**Panels: **heteroskedastic**Correlation: **common AR(1) coefficient for all panels (0.5297)**

|                            |   |    |                  |   |          |
|----------------------------|---|----|------------------|---|----------|
| Estimated covariances      | = | 27 | Number of obs    | = | 376      |
| Estimated autocorrelations | = | 1  | Number of groups | = | 27       |
| Estimated coefficients     | = | 11 | Obs per group:   |   |          |
|                            |   |    | min              | = | 13       |
|                            |   |    | avg              | = | 13.92593 |
|                            |   |    | max              | = | 14       |
|                            |   |    | Wald chi2(10)    | = | 609.89   |
|                            |   |    | Prob > chi2      | = | 0.0000   |

| FSR    | Coefficient | Std. err. | z      | P> z  | [95% conf. interval] |           |
|--------|-------------|-----------|--------|-------|----------------------|-----------|
| lnSIZE | -.1272723   | .0100105  | -12.71 | 0.000 | -.1468926            | -.107652  |
| ROA    | 7.781971    | .7135558  | 10.91  | 0.000 | 6.383427             | 9.180515  |
| CAR    | -.1222964   | .1860773  | -0.66  | 0.511 | -.4870012            | .2424084  |
| NPL    | -.3093238   | .170408   | -1.82  | 0.069 | -.6433174            | .0246698  |
| LIQ    | .4958366    | .145792   | 3.40   | 0.001 | .2100895             | .7815836  |
| lnINC  | .0941865    | .0057596  | 16.35  | 0.000 | .0828978             | .1054752  |
| LTD    | -.0259047   | .0357752  | -0.72  | 0.469 | -.0960228            | .0442133  |
| BI     | -.242235    | .0670478  | -3.61  | 0.000 | -.3736462            | -.1108237 |
| GDP    | .5773549    | .2443018  | 2.36   | 0.018 | .098532              | 1.056178  |
| INF    | -.4424776   | .1182855  | -3.74  | 0.000 | -.6743128            | -.2106423 |
| _cons  | 1.775196    | .1753382  | 10.12  | 0.000 | 1.43154              | 2.118853  |

## Appendix 13: Endogeneity Test Results

```
. ivregress 2sls FSR (lnSIZE=1.lnSIZE) ROA CAR NPL LIQ lnINC LTD BI GDP INF
```

```
Instrumental variables 2SLS regression      Number of obs   =      349
                                           Wald chi2(10)    =     465.38
                                           Prob > chi2       =     0.0000
                                           R-squared         =     0.5685
                                           Root MSE         =     .10244
```

| FSR    | Coefficient | Std. err. | z     | P> z  | [95% conf. interval] |           |
|--------|-------------|-----------|-------|-------|----------------------|-----------|
| lnSIZE | -.0833218   | .0107972  | -7.72 | 0.000 | -.104484             | -.0621596 |
| ROA    | 6.645378    | .7018803  | 9.47  | 0.000 | 5.269718             | 8.021038  |
| CAR    | -.0005532   | .1742966  | -0.00 | 0.997 | -.3421683            | .3410618  |
| NPL    | -.2668655   | .1936809  | -1.38 | 0.168 | -.6464731            | .112742   |
| LIQ    | .6686685    | .1499489  | 4.46  | 0.000 | .3747739             | .962563   |
| lnINC  | .0755792    | .0065785  | 11.49 | 0.000 | .0626856             | .0884727  |
| LTD    | .0275728    | .0355928  | 0.77  | 0.439 | -.0421877            | .0973333  |
| BI     | -.0046151   | .0783863  | -0.06 | 0.953 | -.1582493            | .1490192  |
| GDP    | .0920828    | .3470223  | 0.27  | 0.791 | -.5880685            | .772234   |
| INF    | .0695417    | .1420987  | 0.49  | 0.625 | -.2089665            | .34805    |
| _cons  | .9571256    | .1885579  | 5.08  | 0.000 | .5875589             | 1.326692  |

Instrumented: lnSIZE

Instruments: ROA CAR NPL LIQ lnINC LTD BI GDP INF L.lnSIZE

```
. estat endog
```

Tests of endogeneity

H0: Variables are exogenous

Durbin (score) chi2(1) = 5.37151 (p = 0.0205)

Wu-Hausman F(1,337) = 5.2679 (p = 0.0223)

```
. ivregress 2sls FSR (ROA=1.ROA) lnSIZE CAR NPL LIQ lnINC LTD BI GDP INF
```

```
Instrumental variables 2SLS regression      Number of obs   =      349
                                           Wald chi2(10)    =     363.29
                                           Prob > chi2       =     0.0000
                                           R-squared         =     0.4884
                                           Root MSE         =     .11155
```

| FSR    | Coefficient | Std. err. | z     | P> z  | [95% conf. interval] |           |
|--------|-------------|-----------|-------|-------|----------------------|-----------|
| ROA    | 12.28369    | 1.733975  | 7.08  | 0.000 | 8.88516              | 15.68222  |
| lnSIZE | -.0828141   | .0114989  | -7.20 | 0.000 | -.1053516            | -.0602767 |
| CAR    | -.1498732   | .1960559  | -0.76 | 0.445 | -.5341357            | .2343893  |
| NPL    | -.0998126   | .2150646  | -0.46 | 0.643 | -.5213315            | .3217063  |
| LIQ    | .5884091    | .1645303  | 3.58  | 0.000 | .2659356             | .9108826  |
| lnINC  | .0658168    | .0072679  | 9.06  | 0.000 | .051572              | .0800617  |
| LTD    | -.0238573   | .0412451  | -0.58 | 0.563 | -.1046961            | .0569815  |
| BI     | -.0292853   | .0857379  | -0.34 | 0.733 | -.1973285            | .1387578  |
| GDP    | .2608619    | .3809391  | 0.68  | 0.493 | -.485765             | 1.007489  |
| INF    | -.0304655   | .1575699  | -0.19 | 0.847 | -.3392969            | .2783658  |
| _cons  | 1.169815    | .2171794  | 5.39  | 0.000 | .7441514             | 1.595479  |

Instrumented: ROA

Instruments: lnSIZE CAR NPL LIQ lnINC LTD BI GDP INF L.ROA

```
. estat endog
```

Tests of endogeneity

H0: Variables are exogenous

Durbin (score) chi2(1) = 15.7298 (p = 0.0001)

Wu-Hausman F(1,337) = 15.9059 (p = 0.0001)

```
. ivregress 2sls FSR (CAR=1.CAR) lnSIZE ROA NPL LIQ lnINC LTD BI GDP INF
```

```
Instrumental variables 2SLS regression      Number of obs   =      349
                                           Wald chi2(10)    =     460.58
                                           Prob > chi2      =     0.0000
                                           R-squared       =     0.5689
                                           Root MSE      =     .10239
```

| FSR    | Coefficient | Std. err. | z     | P> z  | [95% conf. interval] |           |
|--------|-------------|-----------|-------|-------|----------------------|-----------|
| CAR    | .0258803    | .2423656  | 0.11  | 0.915 | -.4491476            | .5009082  |
| lnSIZE | -.0776858   | .0114428  | -6.79 | 0.000 | -.1001133            | -.0552582 |
| ROA    | 6.62883     | .7080142  | 9.36  | 0.000 | 5.241148             | 8.016513  |
| NPL    | -.2551752   | .1936058  | -1.32 | 0.187 | -.6346355            | .1242851  |
| LIQ    | .6639513    | .1502543  | 4.42  | 0.000 | .3694584             | .9584443  |
| lnINC  | .0726976    | .0065037  | 11.18 | 0.000 | .0599506             | .0854446  |
| LTD    | .0277655    | .0356535  | 0.78  | 0.436 | -.0421141            | .0976452  |
| BI     | .0020279    | .078382   | 0.03  | 0.979 | -.1515979            | .1556538  |
| GDP    | .0882198    | .3477546  | 0.25  | 0.800 | -.5933667            | .7698064  |
| INF    | .0815154    | .1435449  | 0.57  | 0.570 | -.1998273            | .3628582  |
| _cons  | .891406     | .2033341  | 4.38  | 0.000 | .4928786             | 1.289933  |

Instrumented: CAR

Instruments: lnSIZE ROA NPL LIQ lnINC LTD BI GDP INF L.CAR

```
. estat endog
```

Tests of endogeneity  
H0: Variables are exogenous

```
Durbin (score) chi2(1)      = .013786 (p = 0.9065)
Wu-Hausman F(1,337)       = .013312 (p = 0.9082)
```

```
. ivregress 2sls FSR (NPL=1.NPL) lnSIZE ROA CAR LIQ lnINC LTD BI GDP INF
```

```
Instrumental variables 2SLS regression      Number of obs   =      349
                                           Wald chi2(10)    =     458.84
                                           Prob > chi2      =     0.0000
                                           R-squared       =     0.5679
                                           Root MSE      =     .10251
```

| FSR    | Coefficient | Std. err. | z     | P> z  | [95% conf. interval] |           |
|--------|-------------|-----------|-------|-------|----------------------|-----------|
| NPL    | -.4291917   | .4337138  | -0.99 | 0.322 | -1.279255            | .4208718  |
| lnSIZE | -.078163    | .0107128  | -7.30 | 0.000 | -.0991597            | -.0571663 |
| ROA    | 6.554869    | .7159012  | 9.16  | 0.000 | 5.151728             | 7.958009  |
| CAR    | .0408898    | .1736026  | 0.24  | 0.814 | -.2993651            | .3811447  |
| LIQ    | .6470373    | .1539891  | 4.20  | 0.000 | .3452243             | .9488503  |
| lnINC  | .073253     | .0066175  | 11.07 | 0.000 | .060283              | .086223   |
| LTD    | .0284324    | .035678   | 0.80  | 0.426 | -.0414953            | .0983601  |
| BI     | .0042523    | .0784778  | 0.05  | 0.957 | -.1495613            | .158066   |
| GDP    | .1000209    | .3487841  | 0.29  | 0.774 | -.5835834            | .7836253  |
| INF    | .0840234    | .1425669  | 0.59  | 0.556 | -.1954025            | .3634493  |
| _cons  | .9091838    | .1956391  | 4.65  | 0.000 | .5257383             | 1.292629  |

Instrumented: NPL

Instruments: lnSIZE ROA CAR LIQ lnINC LTD BI GDP INF L.NPL

```
. estat endog
```

Tests of endogeneity  
H0: Variables are exogenous

```
Durbin (score) chi2(1)      = .203204 (p = 0.6521)
Wu-Hausman F(1,337)       = .196332 (p = 0.6580)
```

```
. ivregress 2sls FSR (LIQ=1.LIQ) lnSIZE ROA CAR NPL lnINC LTD BI GDP INF
```

```
Instrumental variables 2SLS regression      Number of obs   =      349
                                           Wald chi2(10)    =     461.44
                                           Prob > chi2      =     0.0000
                                           R-squared       =     0.5609
                                           Root MSE       =     .10334
```

| FSR    | Coefficient | Std. err. | z     | P> z  | [95% conf. interval] |           |
|--------|-------------|-----------|-------|-------|----------------------|-----------|
| LIQ    | 1.044573    | .1960155  | 5.33  | 0.000 | .6603892             | 1.428756  |
| lnSIZE | -.0789464   | .0105704  | -7.47 | 0.000 | -.099664             | -.0582289 |
| ROA    | 6.507869    | .7088281  | 9.18  | 0.000 | 5.118592             | 7.897147  |
| CAR    | .0129176    | .1749901  | 0.07  | 0.941 | -.3300567            | .3558919  |
| NPL    | -.1975099   | .1961862  | -1.01 | 0.314 | -.5820278            | .187008   |
| lnINC  | .0705801    | .0065411  | 10.79 | 0.000 | .0577598             | .0834005  |
| LTD    | .0029995    | .036784   | 0.08  | 0.935 | -.0690958            | .0750948  |
| BI     | .0223161    | .0792775  | 0.28  | 0.778 | -.1330649            | .1776971  |
| GDP    | .0492046    | .3502468  | 0.14  | 0.888 | -.6372665            | .7356757  |
| INF    | .1615007    | .145792   | 1.11  | 0.268 | -.1242464            | .4472478  |
| _cons  | .6111268    | .2071283  | 2.95  | 0.003 | .2051627             | 1.017091  |

Instrumented: LIQ

Instruments: lnSIZE ROA CAR NPL lnINC LTD BI GDP INF L.LIQ

```
. estat endog
```

Tests of endogeneity

H0: Variables are exogenous

Durbin (score) chi2(1) = 9.555 (p = 0.0020)

Wu-Hausman F(1,337) = 9.48618 (p = 0.0022)

```
. ivregress 2sls FSR (lnINC=1.lnINC) lnSIZE ROA CAR NPL LIQ LTD BI GDP INF
```

```
Instrumental variables 2SLS regression      Number of obs   =      349
                                           Wald chi2(10)    =     416.98
                                           Prob > chi2      =     0.0000
                                           R-squared       =     0.5689
                                           Root MSE       =     .10239
```

| FSR    | Coefficient | Std. err. | z     | P> z  | [95% conf. interval] |           |
|--------|-------------|-----------|-------|-------|----------------------|-----------|
| lnINC  | .0738216    | .0080972  | 9.12  | 0.000 | .0579513             | .0896919  |
| lnSIZE | -.0786921   | .0121631  | -6.47 | 0.000 | -.1025312            | -.0548529 |
| ROA    | 6.600201    | .7048826  | 9.36  | 0.000 | 5.218657             | 7.981746  |
| CAR    | .0414005    | .1739609  | 0.24  | 0.812 | -.2995566            | .3823577  |
| NPL    | -.2586051   | .194225   | -1.33 | 0.183 | -.6392792            | .122069   |
| LIQ    | .6591737    | .1505095  | 4.38  | 0.000 | .3641805             | .9541668  |
| LTD    | .0268138    | .0356792  | 0.75  | 0.452 | -.0431162            | .0967438  |
| BI     | .002244     | .0782968  | 0.03  | 0.977 | -.151215             | .155703   |
| GDP    | .0885902    | .3471094  | 0.26  | 0.799 | -.5917317            | .7689121  |
| INF    | .0774883    | .1421094  | 0.55  | 0.586 | -.2010411            | .3560176  |
| _cons  | .8979922    | .1968086  | 4.56  | 0.000 | .5122545             | 1.28373   |

Instrumented: lnINC

Instruments: lnSIZE ROA CAR NPL LIQ LTD BI GDP INF L.lnINC

```
. estat endog
```

Tests of endogeneity

H0: Variables are exogenous

Durbin (score) chi2(1) = .062422 (p = 0.8027)

Wu-Hausman F(1,337) = .060286 (p = 0.8062)

```
. ivregress 2sls FSR (LTD=1.LTD) lnSIZE ROA CAR NPL LIQ lnINC BI GDP INF
```

```
Instrumental variables 2SLS regression      Number of obs   =      349
                                           Wald chi2(10)    =     460.55
                                           Prob > chi2      =     0.0000
                                           R-squared       =     0.5688
                                           Root MSE       =     .1024
```

|        | FSR | Coefficient | Std. err. | z     | P> z  | [95% conf. interval] |
|--------|-----|-------------|-----------|-------|-------|----------------------|
| LTD    |     | .038365     | .0488508  | 0.79  | 0.432 | -.0573808 .1341109   |
| lnSIZE |     | -.077153    | .0104585  | -7.38 | 0.000 | -.0976514 -.0566547  |
| ROA    |     | 6.579213    | .7113623  | 9.25  | 0.000 | 5.184968 7.973457    |
| CAR    |     | .0422284    | .1734298  | 0.24  | 0.808 | -.2976877 .3821445   |
| NPL    |     | -.2562126   | .19361    | -1.32 | 0.186 | -.6356812 .1232559   |
| LIQ    |     | .6502906    | .154635   | 4.21  | 0.000 | .3472115 .9533697    |
| lnINC  |     | .0724009    | .0064775  | 11.18 | 0.000 | .0597053 .0850966    |
| BI     |     | -.0021348   | .0795737  | -0.03 | 0.979 | -.1580963 .1538268   |
| GDP    |     | .0910845    | .3473448  | 0.26  | 0.793 | -.5896988 .7718677   |
| INF    |     | .0778333    | .1420351  | 0.55  | 0.584 | -.2005504 .356217    |
| _cons  |     | .887877     | .1866368  | 4.76  | 0.000 | .5220756 1.253678    |

Instrumented: LTD

Instruments: lnSIZE ROA CAR NPL LIQ lnINC BI GDP INF L.LTD

```
. estat endog
```

Tests of endogeneity  
H0: Variables are exogenous

Durbin (score) chi2(1) = .105599 (p = 0.7452)  
Wu-Hausman F(1,337) = .101999 (p = 0.7496)

```
. ivregress 2sls FSR (BI=1.BI) lnSIZE ROA CAR NPL LIQ lnINC LTD GDP INF
```

```
Instrumental variables 2SLS regression      Number of obs   =      349
                                           Wald chi2(10)    =     458.36
                                           Prob > chi2      =     0.0000
                                           R-squared       =     0.5656
                                           Root MSE       =     .10278
```

|        | FSR | Coefficient | Std. err. | z     | P> z  | [95% conf. interval] |
|--------|-----|-------------|-----------|-------|-------|----------------------|
| BI     |     | .1311306    | .117603   | 1.12  | 0.265 | -.0993671 .3616283   |
| lnSIZE |     | -.0745069   | .0106491  | -7.00 | 0.000 | -.0953787 -.0536351  |
| ROA    |     | 6.559635    | .7052346  | 9.30  | 0.000 | 5.177401 7.94187     |
| CAR    |     | .0599621    | .1739923  | 0.34  | 0.730 | -.2810566 .4009808   |
| NPL    |     | -.2624635   | .1943334  | -1.35 | 0.177 | -.64335 .1184229     |
| LIQ    |     | .687152     | .151349   | 4.54  | 0.000 | .3905134 .9837907    |
| lnINC  |     | .0724321    | .006474   | 11.19 | 0.000 | .0597433 .085121     |
| LTD    |     | .016225     | .0365245  | 0.44  | 0.657 | -.0553617 .0878118   |
| GDP    |     | .1007068    | .3483339  | 0.29  | 0.772 | -.5820151 .7834288   |
| INF    |     | .1551899    | .1516362  | 1.02  | 0.306 | -.1420117 .4523915   |
| _cons  |     | .7972606    | .1949856  | 4.09  | 0.000 | .4150958 1.179425    |

Instrumented: BI

Instruments: lnSIZE ROA CAR NPL LIQ lnINC LTD GDP INF L.BI

```
. estat endog
```

Tests of endogeneity  
H0: Variables are exogenous

Durbin (score) chi2(1) = 2.1792 (p = 0.1399)  
Wu-Hausman F(1,337) = 2.11749 (p = 0.1466)

```
. ivregress 2sls FSR (GDP=1.GDP) lnSIZE ROA CAR NPL LIQ lnINC LTD BI INF
```

```
Instrumental variables 2SLS regression      Number of obs   =      349
                                           Wald chi2(10)   =     49.09
                                           Prob > chi2     =     0.0000
                                           R-squared      =      .
                                           Root MSE      =     .31895
```

| FSR    | Coefficient | Std. err. | z     | P> z  | [95% conf. interval] |          |
|--------|-------------|-----------|-------|-------|----------------------|----------|
| GDP    | -19.03083   | 14.92475  | -1.28 | 0.202 | -48.28281            | 10.22115 |
| lnSIZE | -.0579751   | .035831   | -1.62 | 0.106 | -.1282026            | .0122524 |
| ROA    | 4.194591    | 2.886892  | 1.45  | 0.146 | -1.463614            | 9.852795 |
| CAR    | .751663     | .7698971  | 0.98  | 0.329 | -.7573077            | 2.260634 |
| NPL    | .2480849    | .7186994  | 0.35  | 0.730 | -1.16054             | 1.65671  |
| LIQ    | .9996001    | .5354885  | 1.87  | 0.062 | -.0499381            | 2.049138 |
| lnINC  | .0546412    | .0244745  | 2.23  | 0.026 | .006672              | .1026104 |
| LTD    | -.0801181   | .1389336  | -0.58 | 0.564 | -.3524229            | .1921867 |
| BI     | -.1144212   | .2603103  | -0.44 | 0.660 | -.6246201            | .3957776 |
| INF    | -.0118621   | .4478795  | -0.03 | 0.979 | -.8896897            | .8659656 |
| _cons  | 1.641162    | .8271404  | 1.98  | 0.047 | .0199961             | 3.262327 |

Instrumented: GDP

Instruments: lnSIZE ROA CAR NPL LIQ lnINC LTD BI INF L.GDP

```
. estat endog
```

Tests of endogeneity  
H0: Variables are exogenous

Durbin (score) chi2(1) = 16.0037 (p = 0.0001)  
Wu-Hausman F(1,337) = 16.1962 (p = 0.0001)

```
. ivregress 2sls FSR (INF=1.INF) lnSIZE ROA CAR NPL LIQ lnINC LTD BI GDP
```

```
Instrumental variables 2SLS regression      Number of obs   =      349
                                           Wald chi2(10)   =     458.31
                                           Prob > chi2     =     0.0000
                                           R-squared      =     0.5669
                                           Root MSE      =     .10263
```

| FSR    | Coefficient | Std. err. | z     | P> z  | [95% conf. interval] |           |
|--------|-------------|-----------|-------|-------|----------------------|-----------|
| INF    | -.1032508   | .2876707  | -0.36 | 0.720 | -.6670749            | .4605733  |
| lnSIZE | -.0786577   | .0106864  | -7.36 | 0.000 | -.0996026            | -.0577129 |
| ROA    | 6.70351     | .7129028  | 9.40  | 0.000 | 5.306246             | 8.100774  |
| CAR    | .0796367    | .1795659  | 0.44  | 0.657 | -.2723059            | .4315793  |
| NPL    | -.2447898   | .194421   | -1.26 | 0.208 | -.6258479            | .1362682  |
| LIQ    | .6188091    | .1618018  | 3.82  | 0.000 | .3016834             | .9359348  |
| lnINC  | .0730709    | .0064958  | 11.25 | 0.000 | .0603394             | .0858025  |
| LTD    | .0287448    | .0357004  | 0.81  | 0.421 | -.0412267            | .0987164  |
| BI     | -.0303368   | .0904622  | -0.34 | 0.737 | -.2076396            | .1469659  |
| GDP    | .0800925    | .3477268  | 0.23  | 0.818 | -.6014395            | .7616245  |
| _cons  | .9513942    | .2092088  | 4.55  | 0.000 | .5413525             | 1.361436  |

Instrumented: INF

Instruments: lnSIZE ROA CAR NPL LIQ lnINC LTD BI GDP L.INF

```
. estat endog
```

Tests of endogeneity  
H0: Variables are exogenous

Durbin (score) chi2(1) = .534078 (p = 0.4649)  
Wu-Hausman F(1,337) = .516504 (p = 0.4728)

## Appendix 14: SGMM regression model

Dynamic panel-data estimation, two-step system GMM

|                                   |                    |   |             |
|-----------------------------------|--------------------|---|-------------|
| Group variable: <b>MACK1</b>      | Number of obs      | = | <b>214</b>  |
| Time variable : <b>NAM</b>        | Number of groups   | = | <b>27</b>   |
| Number of instruments = <b>27</b> | Obs per group: min | = | <b>7</b>    |
| F(11, 26) = <b>224403.63</b>      | avg                | = | <b>7.93</b> |
| Prob > F = <b>0.000</b>           | max                | = | <b>8</b>    |

| FSS    | Coefficient      | Std. err.       | t            | P> t         | [95% conf. interval] |                  |
|--------|------------------|-----------------|--------------|--------------|----------------------|------------------|
| FSS    |                  |                 |              |              |                      |                  |
| L1.    | <b>.407201</b>   | <b>.0988127</b> | <b>4.12</b>  | <b>0.000</b> | <b>.2040887</b>      | <b>.6103134</b>  |
| lnSIZE | <b>-.0715745</b> | <b>.0150798</b> | <b>-4.75</b> | <b>0.000</b> | <b>-.1025714</b>     | <b>-.0405776</b> |
| ROA    | <b>5.359878</b>  | <b>1.586014</b> | <b>3.38</b>  | <b>0.002</b> | <b>2.09978</b>       | <b>8.619976</b>  |
| CAR    | <b>-.1048387</b> | <b>.230762</b>  | <b>-0.45</b> | <b>0.653</b> | <b>-.5791768</b>     | <b>.3694995</b>  |
| NPL    | <b>-.9797766</b> | <b>.5344733</b> | <b>-1.83</b> | <b>0.078</b> | <b>-2.078402</b>     | <b>.1188491</b>  |
| LIQ    | <b>-.9407358</b> | <b>.618676</b>  | <b>-1.52</b> | <b>0.140</b> | <b>-2.212442</b>     | <b>.3309709</b>  |
| lnINC  | <b>.0650332</b>  | <b>.0167124</b> | <b>3.89</b>  | <b>0.001</b> | <b>.0306803</b>      | <b>.099386</b>   |
| LTD    | <b>.1775695</b>  | <b>.0796277</b> | <b>2.23</b>  | <b>0.035</b> | <b>.0138924</b>      | <b>.3412467</b>  |
| BI     | <b>.2647368</b>  | <b>.1525323</b> | <b>1.74</b>  | <b>0.094</b> | <b>-.0487979</b>     | <b>.5782714</b>  |
| GDP    | <b>.4732932</b>  | <b>.1935642</b> | <b>2.45</b>  | <b>0.022</b> | <b>.0754163</b>      | <b>.8711701</b>  |
| INF    | <b>-2.159927</b> | <b>.5955718</b> | <b>-3.63</b> | <b>0.001</b> | <b>-3.384142</b>     | <b>-.9357118</b> |
| _cons  | <b>1.822373</b>  | <b>.6576729</b> | <b>2.77</b>  | <b>0.010</b> | <b>.4705066</b>      | <b>3.174239</b>  |

Warning: Uncorrected two-step standard errors are unreliable.

Warning: Uncorrected two-step standard errors are unreliable.

Instruments for first differences equation

Standard

D.(L3.CAR L6.NPL L3.lnINC L6.LTD L3.BI L6.INF)

GMM-type (missing=0, separate instruments for each period unless collapsed)

L(2/5).(L3.lnSIZE L6.ROA L6.LIQ L3.GDP) collapsed

Instruments for levels equation

Standard

L3.CAR L6.NPL L3.lnINC L6.LTD L3.BI L6.INF

\_cons

GMM-type (missing=0, separate instruments for each period unless collapsed)

DL.(L3.lnSIZE L6.ROA L6.LIQ L3.GDP) collapsed

Arellano-Bond test for AR(1) in first differences: z = **-1.81** Pr > z = **0.070**  
 Arellano-Bond test for AR(2) in first differences: z = **-0.46** Pr > z = **0.649**

Sargan test of overid. restrictions: chi2(15) = **22.06** Prob > chi2 = **0.106**  
 (Not robust, but not weakened by many instruments.)

Hansen test of overid. restrictions: chi2(15) = **22.22** Prob > chi2 = **0.102**  
 (Robust, but weakened by many instruments.)

Difference-in-Hansen tests of exogeneity of instrument subsets:

GMM instruments for levels

Hansen test excluding group: chi2(11) = **14.44** Prob > chi2 = **0.209**

Difference (null H = exogenous): chi2(4) = **7.77** Prob > chi2 = **0.100**

iv(L3.CAR L6.NPL L3.lnINC L6.LTD L3.BI L6.INF)

Hansen test excluding group: chi2(9) = **19.02** Prob > chi2 = **0.025**

Difference (null H = exogenous): chi2(6) = **3.20** Prob > chi2 = **0.784**

.

**Appendix 15: Quantile Regression at Different Percentiles**

|        | (1)<br>Q10                | (2)<br>Q25                | (3)<br>Q50                   | (4)<br>Q75                  | (5)<br>Q90                  |
|--------|---------------------------|---------------------------|------------------------------|-----------------------------|-----------------------------|
| lnSIZE | <b>-0.0300</b><br>[-1.09] | <b>-0.0317</b><br>[-1.10] | <b>-0.0890***</b><br>[-4.00] | <b>-0.106***</b><br>[-9.57] | <b>-0.128***</b><br>[-9.92] |
| ROA    | <b>8.419**</b><br>[2.55]  | <b>12.69***</b><br>[3.68] | <b>11.16***</b><br>[4.58]    | <b>9.810***</b><br>[4.13]   | <b>10.88***</b><br>[5.18]   |
| CAR    | <b>-0.181</b><br>[-0.89]  | <b>-0.277</b><br>[-1.35]  | <b>-0.143</b><br>[-0.56]     | <b>0.391**</b><br>[1.97]    | <b>0.199</b><br>[0.49]      |
| NPL    | <b>-0.142</b><br>[-0.22]  | <b>-0.337</b><br>[-0.86]  | <b>-0.0879</b><br>[-0.18]    | <b>-0.210</b><br>[-0.83]    | <b>-0.406</b><br>[-0.85]    |
| LIQ    | <b>0.476**</b><br>[2.16]  | <b>0.385**</b><br>[2.41]  | <b>0.705***</b><br>[3.25]    | <b>0.662***</b><br>[3.25]   | <b>0.535*</b><br>[1.81]     |
| lnINC  | <b>0.0267</b><br>[1.41]   | <b>0.0227</b><br>[1.13]   | <b>0.0614***</b><br>[3.90]   | <b>0.101***</b><br>[16.33]  | <b>0.113***</b><br>[10.26]  |
| LTD    | <b>0.0124</b><br>[0.28]   | <b>-0.0189</b><br>[-0.34] | <b>-0.0167</b><br>[-0.32]    | <b>-0.195***</b><br>[-2.65] | <b>-0.0953</b><br>[-1.03]   |
| BI     | <b>0.0506</b><br>[0.86]   | <b>-0.0716</b><br>[-0.99] | <b>-0.179</b><br>[-1.47]     | <b>-0.0846</b><br>[-0.84]   | <b>-0.199**</b><br>[-2.50]  |
| GDP    | <b>0.0679</b><br>[0.24]   | <b>0.246</b><br>[0.79]    | <b>0.164</b><br>[0.62]       | <b>0.469</b><br>[0.88]      | <b>0.459</b><br>[0.76]      |
| INF    | <b>0.0640</b><br>[0.48]   | <b>-0.104</b><br>[-0.57]  | <b>0.0480</b><br>[0.18]      | <b>0.169</b><br>[0.63]      | <b>0.333</b><br>[1.06]      |
| _cons  | <b>0.745***</b><br>[3.36] | <b>0.978***</b><br>[4.01] | <b>1.259***</b><br>[5.64]    | <b>1.218***</b><br>[4.55]   | <b>1.586***</b><br>[6.25]   |
| N      | <b>378</b>                | <b>378</b>                | <b>378</b>                   | <b>378</b>                  | <b>378</b>                  |

t statistics in brackets

\* p&lt;0.1, \*\* p&lt;0.05, \*\*\* p&lt;0.01